

Turán densities of complete 4-graphs via flag algebras on seven vertices

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8 October 2026

Abstract

For $r \geq 3$ the Turán density $\pi(K_t^{(r)})$ of the complete r -graph is not known for any $t > r$. Among 4-graphs the only case with a plausible conjecture is $K_5^{(4)}$: Giraud's construction gives $\pi(K_5^{(4)}) \geq 11/16$, and the best published upper bound is 0.73628. We prove $\pi(K_5^{(4)}) < 0.69160$, which removes 91.6% of the gap. We also determine the codegree-squared density $\sigma(K_5^{(4)}) = 31/64$, as conjectured by Balogh, Clemen and Lidický, with a stability theorem: near-extremal $K_5^{(4)}$ -free graphs are close to complements of balanced Giraud systems. We prove $\pi(K_6^{(4)}) < 0.86018$ and $\pi(K_7^{(4)}) < 0.93524$ (previously about 0.8946 and 0.9444), $\pi(K_5^{4-}) < 0.44701$ and $\pi(K_6^{4-}) < 0.78055$ for $K_5^{(4)}$ and $K_6^{(4)}$ minus an edge, and bounds for three 5-graphs and nine threshold families of 4-graphs. We improve the codegree-squared bounds for six 3-graphs, among them J_4 and $K_5^<$, give such bounds for four further 4-graphs, and show that the value conjectured for $\sigma(J_4)$ rests on an arithmetic slip. Each bound is a plain flag-algebra certificate on seven vertices with exact rational entries; its value is the exact minimum of an explicit function over all admissible 7-vertex graphs, computed by exhaustive search in two independent programs. On six vertices the same method gives exactly de Caen's bounds for $K_5^{(4)}$ and $K_6^{(4)}$ and nothing for $K_7^{(4)}$, which may explain why we found no earlier flag-algebra bound on the Turán density of a complete 4-graph. An exact dual point shows that no plain seven-vertex certificate can improve our bound for $K_5^{(4)}$ by more than $2.2 \cdot 10^{-7}$. Consequences are asymptotic lower bounds for lottery numbers and covering insertion codes; the quadruples of every Giraud-type construction cost at least $7/16$ to cover fractionally by k -sets, sharp for $k = 6$.

1 Introduction

For an r -graph F , the Turán number $\text{ex}(n, F)$ is the largest number of edges of an n -vertex r -graph with no copy of F , and the Turán density is $\pi(F) = \lim_{n \rightarrow \infty} \text{ex}(n, F) / \binom{n}{r}$. Turán [37] determined $\pi(K_t^{(2)}) = 1 - 1/(t - 1)$. For $r \geq 3$ no value $\pi(K_t^{(r)})$ with $t > r$ is known; see [10] and the surveys [33, 20]. For the tetrahedron $K_4^{(3)}$, whose conjectured density is $5/9$, Jeong, Park, Im, Lee and Yang [17] recently proved the upper bound 0.557789 (arXiv v2), using a seven-vertex flag-algebra certificate found by cutting planes and column generation and checked in Lean. We use their search scheme in an independent implementation.

The 4-uniform case. For 4-graphs the analogous problem is $K_5^{(4)}$. Giraud [14] constructed 4-graphs of density $5/16 + o(1)$ in which every five vertices span an edge (Definition 3); their complements are $K_5^{(4)}$ -free of density $11/16 + o(1)$. Sidorenko conjectured that $11/16$ is the answer [33, Conjecture 7].

Main results. Theorem 2 improves the Turán bounds for $K_5^{(4)}$, $K_6^{(4)}$, $K_7^{(4)}$, K_5^{4-} and K_6^{4-} ; Table 1 compares them with the literature. Our second main result is $\sigma(K_5^{(4)}) = 31/64$ (Theorem 4), proving the conjecture of Balogh, Clemen and Lidický [2, Conjecture 8.2 of arXiv v3]. The certificate is tight on the seven-vertex subgraphs of Giraud’s construction. This leads to stability with linear dependence and no removal lemma (Theorem 5), using a characterisation of Giraud systems by their seven-vertex subgraphs (Theorem 13).

Theorems 21 and 24 delimit the method: plain flag-algebra certificates on six vertices cannot improve the classical bounds for $K_5^{(4)}$, $K_6^{(4)}$ and K_5^{4-} , and our seven-vertex bound for $K_5^{(4)}$ is within $2.2 \cdot 10^{-7}$ of the best plain seven-vertex bound.

Further results and applications. The same method gives three 5-graph bounds (Theorem 26), nine threshold-family bounds (Theorem 27), and upper bounds for ten further codegree-squared densities (Table 7; for the four 4-graphs among them we found no published bound). Section 6.3 also shows that the value 0.28 conjectured for J_4 in [2] rests on an arithmetic slip: the construction gives $16/57$, and the plain seven-vertex method stops at least $1.5 \cdot 10^{-6}$ above it (Remark 31).

The applications are lower bounds for lottery numbers and covering insertion codes (Corollaries 34 and 45). Theorem 35, with a short human proof, shows that Giraud systems are expensive to cover. Constructions and Kahn’s theorem [18] give three upper bounds for asymptotic lottery numbers (Proposition 44); the one for $(6, 4, 5)$ makes Theorem 35 sharp for $k = 6$.

Why seven vertices? Flag algebras [32] have been used for the codegree-squared density of $K_5^{(4)}$ [2] and for tight cycles in 4-graphs [5]; we found no earlier bound on the Turán density of a complete 4-graph by this method. Theorem 21 explains the limitation of plain certificates on six vertices. At seven vertices there are 7013320 isomorphism classes of 4-graphs, and the semidefinite program has blocks of order up to 1024 and several million constraints. The scheme of Jeong et al. [17] makes the search feasible; exact arithmetic certifies the resulting bounds.

Verification. Each flag-algebra bound is given by rational positive semidefinite matrices and an exact minimum over all admissible seven-vertex graphs. Section 3 proves the averaging inequality and describes two independent implementations of the exhaustive check. No proof assistant has checked these certificates.

Organisation. Section 2 states the main results and Section 3 explains the exact certificates. Section 4 proves the codegree-squared result and stability. Section 5 gives the six- and seven-vertex limitations, Section 6 collects further bounds, and Section 7 gives the applications. Section 8 describes the data and verification. Appendices A and B give the remaining exact bounds and notation.

2 Definitions and main results

An r -graph G is a set $V(G)$ of vertices with a set $E(G)$ of r -subsets, the edges; $e(G) = |E(G)|$ and $d(G) = e(G)/\binom{|V(G)|}{r}$. For $S \subseteq V(G)$, $G[S]$ is the induced subgraph.

2.1 Turán densities

Definition 1 (Turán systems). Let $p > r$ and $\lambda \geq 1$. An r -graph G is (p, λ) -admissible if every p -subset of $V(G)$ spans at least λ edges; for $\lambda = 1$ we say p -admissible, and G is then

a *Turán* (n, p, r) -system. Let $T_\lambda(n, p, r)$ be the least number of edges of a (p, λ) -admissible r -graph on n vertices, and $T(n, p, r) = T_1(n, p, r)$.

Admissibility is hereditary: every induced subgraph of an admissible graph is admissible. By averaging over the $(n-1)$ -subsets, $T_\lambda(n, p, r)/\binom{n}{r}$ is non-decreasing in n ([19, Theorem 1 and Corollary 1] for $\lambda = 1$), so the limit $t_\lambda(p, r) = \lim_n T_\lambda(n, p, r)/\binom{n}{r}$ exists; $t(p, r) = t_1(p, r)$ is the *Turán density* in Sidorenko's normalisation [33, 34]. An r -graph is p -admissible exactly when its complement in $K_n^{(r)}$ is $K_p^{(r)}$ -free, so

$$\pi(K_p^{(r)}) = 1 - t(p, r).$$

More generally, let H_k^r be the r -graph on $r+1$ vertices with k edges (it is unique up to isomorphism, since an r -subset of an $(r+1)$ -set is determined by the missing vertex). An r -graph is H_k^r -free iff every $(r+1)$ -set spans at most $k-1$ edges, iff its complement spans at least $r+2-k$ edges in every $(r+1)$ -set. Hence $\pi(H_k^r) = 1 - t_{r+2-k}(r+1, r)$. We write $K_5^{4-} = H_4^4$, the 4-graph on five vertices with four edges; then $\pi(K_5^{4-}) = 1 - t_2(5, 4)$. Also $K_6^{(5)} = H_6^5$. Finally K_6^{4-} is $K_6^{(4)}$ minus one edge: a 4-graph contains K_6^{4-} iff some six vertices span at least 14 edges, so $\pi(K_6^{4-}) = 1 - t_2(6, 4)$.

Theorem 2. (a) $t(5, 4) \geq \frac{35604499940047}{115448720916480} > 0.308400$, hence $\pi(K_5^{(4)}) < 0.691600$.

$$(b) \ t(6, 4) \geq \frac{9224492822869}{65970697666560} > 0.139827, \text{ hence } \pi(K_6^{(4)}) < 0.860173.$$

$$(c) \ t(7, 4) \geq \frac{7476698908057}{115448720916480} > 0.064762, \text{ hence } \pi(K_7^{(4)}) < 0.935238.$$

$$(d) \ t_2(5, 4) \geq \frac{48641756816487}{87960930222080} > 0.552992, \text{ hence } \pi(K_5^{4-}) < 0.447008.$$

$$(e) \ t_2(6, 4) \geq \frac{202684381843379}{923589767331840} > 0.219452, \text{ hence } \pi(K_6^{4-}) < 0.780548.$$

The denominators are $105 \cdot 2^{40}$ in (a) and (c), and $15 \cdot 2^{42}$, $5 \cdot 2^{44}$ and $105 \cdot 2^{43}$ in (b), (d) and (e); they come from the scaling of the integer certificates (Section 3.3).

2.2 Codegree-squared density and stability

Giraud systems.

Definition 3 (Giraud systems). Let $V = A \cup B$ be a partition, one part possibly empty, and M a 0/1 matrix indexed by $A \times B$. The *Giraud system* $\Gamma = \Gamma(A, B, M)$ is the 4-graph on V whose edges are all 4-subsets of A , all 4-subsets of B , and the *cross* 4-sets $\{a, a', b, b'\}$ ($a, a' \in A, b, b' \in B$) whose minor $M_{ab} + M_{ab'} + M_{a'b} + M_{a'b'}$ is even.

Giraud's construction [14] is a Giraud system with a specific matrix and equal parts; it is a *Turán* $(n, 5, 4)$ -system whose complement is $K_5^{(4)}$ -free. With equal parts the internal 4-sets have density $1/8 + o(1)$ and the cross ones $3/8 + o(1)$, so every Giraud system with equal parts in which a fraction $1/2 + o(1)$ of the minors are even, for instance one with a uniformly random matrix, has density $5/16 + o(1)$; we use the name Giraud's construction for these balanced systems too. Definition 3 is the form described by Sidorenko [34, Section 2], with arbitrary M and arbitrary part sizes. At uniformity 4 it is also exactly Construction 2 of [33], the generalisation by de Caen, Kreher and Wiseman [9].

For a 4-graph G and a 3-set $T \subseteq V(G)$ let $\deg_G(T)$ be the number of edges containing T , and let $\text{co}_2(G) = \sum_T \deg_G(T)^2$, the sum over all 3-subsets of $V(G)$. Balogh, Clemen and Lidický [2] define the *codegree-squared density* of a 4-graph F as

$$\sigma(F) = \limsup_{n \rightarrow \infty} \max \left\{ \frac{\text{co}_2(G)}{\binom{n}{3}(n-3)^2} : G \text{ is an } F\text{-free 4-graph on } n \text{ vertices} \right\}.$$

They state $31/64 \leq \sigma(K_5^{(4)}) \leq 0.55241250$, the lower bound by Giraud's construction and the upper bound by flag algebras, and conjecture that the lower bound is tight [2, Conjecture 8.2 of arXiv v3].

Theorem 4. $\sigma(K_5^{(4)}) = 31/64$.

The proof (Section 4.2) is a plain seven-vertex certificate for the objective $\kappa = 2d - \gamma$, a combination of densities on five vertices (quadratic in the codegrees). Unlike every other certificate in this paper, it gives a sharp bound, matched by Giraud's construction; the minimum $33/64$ of $\kappa - \sum c_Q$ is attained exactly, and only, on the ten isomorphism classes of 7-vertex subgraphs of Giraud's construction. That rigidity gives stability: a near-extremal graph can be made into the complement of a balanced Giraud system by changing a small proportion of its 4-sets. The following theorem gives explicit constants.

Theorem 5. *For every $\delta > 0$ there are $\varepsilon > 0$ and n_0 , namely $\varepsilon = \delta/(2C_0)$ and $n_0 = \lceil 7 + 2(69C_0 + 24)/\delta \rceil$ with $C_0 < 7.81 \cdot 10^8$ as in Theorem 17, such that every $K_5^{(4)}$ -free 4-graph G on $n \geq n_0$ vertices with $\text{co}_2(G) \geq (31/64 - \varepsilon)\binom{n}{3}(n-3)^2$ becomes the complement of a Giraud system after changing at most $\delta\binom{n}{4}$ of its 4-sets. The Giraud system can be chosen with parts of sizes $n/2 \pm O(\sqrt{\varepsilon + 1/n})n$ and a fraction $1/2 + O(\varepsilon + 1/n)$ of even minors, that is, with a quasirandom matrix. (The roles of ε and δ are reversed with respect to Hypothesis 40.)*

Section 4 proves an explicit version (Theorem 17) from the margin of the certificate outside Giraud's ten classes and from Theorem 13: a 4-graph on at least seven vertices is a Giraud system iff all its 7-vertex induced subgraphs are.

2.3 Comparison with previous bounds

Table 1 compares Theorem 2 and the 5-graph bounds of Theorem 26 with the literature. We explain the previous bounds here, including those obtained by averaging rather than stated in a source.

	this paper	previous upper bound	lower bound	gap removed
$\pi(K_5^{(4)})$	< 0.691600	$751/1020 < 0.736275$ [34]	$11/16$ [14]	91.6%
$\pi(K_6^{(4)})$	< 0.860173	< 0.894570 (from [34])	> 0.845898 [34]	70.7%
$\pi(K_7^{(4)})$	< 0.935238	< 0.944386 (from [34])	> 0.913476 [34]	29.6%
$\pi(K_5^{4-})$	< 0.447008	$1/2$ [23, 36]	> 0.428571 [36]	74.2%
$\pi(K_6^{4-})$	< 0.780548	< 0.852398 (derivable)	$11/16$ [14]	43.6%
$\pi(K_6^{(5)})$	< 0.767401	< 0.769678 [6]		
$\pi(H_4^5)$	< 0.389709	$2/5$ [23]		
$\pi(K_7^{(5)})$	< 0.927918	$14/15 < 0.933334$ [8]		

Table 1: The new bounds against the literature (upper bounds rounded up, lower bounds rounded down). The last column is the share of the previous interval [lower, upper] removed. Construction bounds for the three 5-graphs were not checked.

For $K_5^{(4)}$, Markström [22] proved $\text{ex}(17, K_5^{(4)}) \leq 1753$. Sidorenko's integer rounding at $n = 18$ gives the bound $751/1020 < 0.736275$ [34, Section 2]. This bound, or Markström's, is quoted as the best one in [2, 29, 13]. Repeated averaging from $T(18, 5, 4) \geq 807$ gives the unpublished 0.735614, against which the gap removed is 91.5%.

For $K_6^{(4)}$ and $K_7^{(4)}$ we know of no conjectured value. The entries "from [34]" in Table 1 use $T(n, p, 4) \geq \lceil nT(n-1, p, 4)/(n-4) \rceil$, starting from $T(16, 6, 4) \geq 190$ and $T(16, 7, 4) \geq 99$ in Sidorenko's tables and stopping at $n = 200\,000$. This gives a valid bound because $T(n, p, 4)/\binom{n}{4}$ is non-decreasing; the same rule is used for K_6^{4-} and in Table 6. The construction bounds are [34, Ex. 3.4] for $K_6^{(4)}$ and $4677/5120$ [34, Prop. 6.1] for $K_7^{(4)}$. For $K_6^{(5)}$, the upper bound is $(43 + \sqrt{345})/80$, stated in [6, Theorem 3] with the proof omitted.

For K_5^{4-} the published bounds $3/7 \leq \pi(K_5^{4-}) \leq 1/2$ are those of Sidorenko [36], the upper one being the general bound $\pi(H_k^r) \leq (k-2)/r$ [23, Cor. 3.9]. The problem is the case $(n; 5, 4)$ of the Brown–Erdős–Sós problem for 4-graphs: a 4-graph is K_5^{4-} -free iff every five vertices span at most three edges.

For K_6^{4-} , the case $(n; 6, 14)$ of the same problem, repeated averaging $T_2(n, 6, 4) \geq \lceil nT_2(n-1, 6, 4)/(n-4) \rceil$ from $T_2(6, 6, 4) = 2$ gives 0.852398. We found no published bound specific to this graph beyond the inclusions $K_5^{(4)} \subset K_6^{4-} \subset K_6^{(4)}$; the first inclusion gives the lower bound.

3 Flag-algebra certificates on seven vertices

The proof has three parts: the averaging inequality, rational positive semidefinite matrices, and an exact minimum over every admissible seven-vertex graph. We describe these checks first, then the search that produced the candidate matrices.

3.1 The inequality

We use the flag-algebra inequality of Razborov [32] in an elementary form, written for 4-graphs and seven vertices; nothing below depends on the general theory.

A *type* of size s is a 4-graph σ on $[s]$. A σ -*flag* on $m \geq s$ vertices is a pair (F, θ) where F is a 4-graph on m vertices and $\theta: [s] \rightarrow V(F)$ is injective with $\theta(X) \in E(F) \Leftrightarrow X \in E(\sigma)$ for every 4-set $X \subseteq [s]$. Two σ -flags are isomorphic if a bijection of the vertex sets maps edges to edges and θ to θ' . Let $\mathcal{F}_\sigma^m$ be a finite list of pairwise non-isomorphic σ -flags on m vertices. For a 4-graph H , an injective $\theta: [s] \rightarrow V(H)$ and a set $U \subseteq V(H) \setminus \theta([s])$ of size $m-s$, write H_θ^U for the pair $(H[\theta([s]) \cup U], \theta)$; it is a σ -flag exactly when $H[\theta] = \sigma$, that is, when θ is an isomorphism from σ onto $H[\theta([s])]$.

Definition 6 (Plain certificate on N vertices). A *block* is a tuple $(s, m, \sigma, \mathcal{F}_\sigma^m, Q)$ with $2m-s \leq N$, a type σ of size s , a flag list $\mathcal{F}_\sigma^m$ and a positive semidefinite matrix Q indexed by $\mathcal{F}_\sigma^m$. For a 4-graph H on N vertices put

$$c_Q(H) = \mathbb{E}_{(\theta, U_1, U_2)} \left[Q(H_\theta^{U_1}, H_\theta^{U_2}) \right],$$

where (θ, U_1, U_2) is uniform among the injective maps $\theta: [s] \rightarrow V(H)$ and the ordered pairs of *disjoint* $(m-s)$ -subsets of $V(H) \setminus \theta([s])$, and where $Q(F, F')$ is the entry of Q if both arguments are σ -flags isomorphic to members of $\mathcal{F}_\sigma^m$, and 0 otherwise. A *plain certificate on N vertices* for (p, λ) is a finite list of blocks; its *value* is

$$b = \min_H \left(d(H) - \sum_{\text{blocks}} c_Q(H) \right),$$

the minimum over all (p, λ) -admissible 4-graphs H on N vertices.

The number of triples (θ, U_1, U_2) in an N -vertex graph is $\text{conf}(s, m) = \frac{N!}{(N-s)!} \binom{N-s}{m-s} \binom{N-m}{m-s}$.

Proposition 7. *If a plain certificate on N vertices for (p, λ) has value b , then $t_\lambda(p, 4) \geq b$.*

Proof. Let G be a (p, λ) -admissible 4-graph on $n \geq N$ vertices and fix one block. For an injective $\theta: [s] \rightarrow V(G)$ with $G[\theta] = \sigma$, let v_θ be the vector indexed by $\mathcal{F}_\sigma^m$ with $v_\theta(F) = \mathbb{P}_U[G_\theta^U \cong F]$, U a uniform $(m-s)$ -subset of $V(G) \setminus \theta([s])$; put $v_\theta = 0$ if $G[\theta] \neq \sigma$. Since $Q \succeq 0$, $v_\theta^\top Q v_\theta \geq 0$, and $v_\theta^\top Q v_\theta = \mathbb{E}_{U_1, U_2} Q(G_\theta^{U_1}, G_\theta^{U_2})$ with U_1, U_2 independent. Conditioning on $U_1 \cap U_2 = \emptyset$ changes this expectation by at most $2 \max |Q| \cdot \mathbb{P}[U_1 \cap U_2 \neq \emptyset] = O(1/n)$. Hence the average over a uniform configuration (θ, U_1, U_2) of G , with U_1, U_2 disjoint, is at least $-O(1/n)$.

A configuration occupies exactly $2m - s \leq N$ vertices. Every set of $2m - s$ vertices lies in the same number of N -subsets, and every N -subset contains $\text{conf}(s, m)$ configurations. So a uniform configuration of G is obtained by choosing a uniform N -subset S and then a uniform configuration of $G[S]$; the flag G_θ^U does not change when G is replaced by $G[S]$. Therefore $\mathbb{E}_S c_Q(G[S]) \geq -O(1/n)$ for each block. Also $d(G) = \mathbb{E}_S d(G[S])$. Each $G[S]$ is admissible, so

$$d(G) = \mathbb{E}_S \left(d(G[S]) - \sum c_Q(G[S]) \right) + \sum \mathbb{E}_S c_Q(G[S]) \geq b - O(1/n).$$

Applying this to extremal graphs and letting $n \rightarrow \infty$ gives $t_\lambda(p, 4) \geq b$. $\square$

No symmetry of Q is assumed, and flags or types missing from the lists only make the inequality weaker; the lists need not be complete for soundness (ours are complete; Section 3.5). Theorems 2 and 26 are Proposition 7 applied to the certificates in the data package [25], with the value b computed exactly as described below.

Remark 8. The proof uses the edge density only through the exact identity $d(G) = \mathbb{E}_S d(G[S])$ for a uniform N -subset S . So it applies verbatim to any objective ψ in place of d that is defined on all 4-graphs with at least N vertices and satisfies $\psi(G) = \mathbb{E}_S \psi(G[S])$ whenever $|V(G)| \geq N$: if the certificate with d replaced by ψ has value b , then $\psi(G) \geq b - O(1/n)$ for every (p, λ) -admissible G on $n \geq N$ vertices. Every induced density of a configuration on at most N vertices has this property, for instance the probability that a uniform sequence of $j \leq N$ distinct vertices spans a given labelled 4-graph, and so has every linear combination of such densities.

3.2 Blocks at seven vertices

At $N = 7$ the blocks with $m > s$ and $m \geq 4$ are $(s, m) \in \{(1, 4), (2, 4), (3, 4), (3, 5), (4, 5), (5, 6)\}$; the other blocks ($m \leq 3$ or $m = s$) have a single flag per type and only lower the bound. Table 2 gives the sizes. A type on $s \leq 3$ vertices has no edge, so these blocks have one type. A 4-graph on five vertices is determined up to isomorphism by its number of edges, so the types of the $(5, 6)$ block are indexed by $e(\sigma) \in \{0, \dots, 5\}$, restricted to the admissible ones. A σ -flag on $m = s + 1$ vertices is determined by the link of the free vertex, a set of triples of $[s]$; for $(5, 6)$ there are $2^{10} = 1024$ links. The $(5, 6)$ block needs $2m - s = 7$, so it does not exist at $N = 6$; Section 5 gives an exact lower bound on its contribution.

3.3 Exact rounding

Let $Q \approx \sum_i w_i u_i u_i^\top$ be the floating-point solution of a block in eigenform. We drop eigenvalues below a small relative threshold, round $a_i = \text{round}(M \sqrt{w_i} u_i)$ to an integer vector, and set $\tilde{Q} = A^\top A / M^2$, where the rows of A are the a_i and $M = 2^{20}$ or 2^{21} . Then $\tilde{Q}$ is rational and positive semidefinite by construction (compare [28]). The certificate is the list of integer matrices A ; the floating-point solution plays no further role. The value b is computed for $\tilde{Q}$ itself, so rounding may change the bound but never invalidates it.

(s, m)	(1, 4)	(2, 4)	(3, 4)	(3, 5)	(4, 5)	(5, 6)
$\text{conf}(s, m)$	140	1260	2520	1260	5040	5040
$t(5, 4)$	2	2	2	23	15, 16	809, 854, 904, 960, 1024
$t(6, 4)$	2	2	2	24	16, 16	1023, 1024 ($\times 5$)
$t(7, 4)$	2	2	2	24	16, 16	1024 ($\times 6$)
$t_2(5, 4)$	2	2	2	19	11, 15	373, 463, 588, 768
$t_2(6, 4)$	2	2	2	24	16, 16	1013, 1023, 1024 ($\times 4$)

Table 2: Number of admissible flags per type at $N = 7$. For (4, 5) the types are the empty and the one-edge 4-graph; for (5, 6) they are the admissible 5-vertex types in increasing number of edges (1 to 5 edges for $t(5, 4)$, 0 to 5 for $t(6, 4)$, $t(7, 4)$ and $t_2(6, 4)$, 2 to 5 for $t_2(5, 4)$).

3.4 Exact evaluation over all admissible graphs

All the numbers $\text{conf}(s, m)$ in Table 2 divide 5040, and $A^\top A$ has integer entries. So for every 7-vertex H the number $Z(H) = 5040 M^2 \sum c_{\tilde{Q}}(H)$ is an integer, and

$$d(H) - \sum c_{\tilde{Q}}(H) = \frac{144 M^2 e(H) - Z(H)}{5040 M^2}.$$

The value b is the minimum of this fraction over the admissible H , computed in integer arithmetic.

Lemma 9. *Let $R_1, \dots, R_q$ represent the isomorphism classes of (p, λ) -admissible 4-graphs on $\{0, \dots, 5\}$. Every (p, λ) -admissible 4-graph on seven vertices is isomorphic to a graph $R_i \cup L$ obtained by adding the vertex 6 with link L , a set of triples of $\{0, \dots, 5\}$.*

Proof. Delete any vertex v of H ; the rest is admissible, hence isomorphic to some R_i . Transport the isomorphism and send v to 6; the link of 6 is the image of the link of v . $\square$

Since $d - \sum c_{\tilde{Q}}$ is an isomorphism invariant, b is the minimum over the admissible graphs $R_i \cup L$, L ranging over all 2^{20} links. We computed b twice in the first implementation: over a list of canonical representatives of all admissible classes, produced by one-vertex extension with nauty canonical forms [24]; and over all admissible raw extensions $R_i \cup L$, which needs no canonical form. Both scans give the same minimum and the same maxima of Z on every group of graphs with equal edge count and equal value of a second invariant. Table 3 gives the sizes. The class counts were also confirmed by the labelled count $\sum_H 7! / |\text{Aut}(H)|$ and, in the second implementation, by Burnside’s lemma over S_7 .

	$t(5, 4)$	$t(6, 4)$	$t(7, 4)$	$t_2(5, 4)$	$t_2(6, 4)$
classes, 6 vertices	122	155	156	62	154
classes, 7 vertices	3 908 438	7 011 184	7 013 319	360 127	6 986 573
labelled, 7 vertices	19 199 206 747	34 352 419 335	$2^{35} - 1$	1 741 143 596	34 244 802 014
raw extensions $R_i \cup L$	86 952 880	162 481 445	163 577 855	10 058 620	160 859 017

Table 3: Sizes of the exhaustive scans (admissible graphs only). For $p = 7$ a 7-vertex graph is admissible unless it is empty; 7 013 320 is the number of 3-graphs on seven unlabelled vertices.

3.5 Independent verification

A second implementation, written separately and sharing no code with the first, re-checked every certificate. Its decisive check, the raw scan, reads only the certificate files (blocks, types, flag lists and the integer matrices A). It differs from the first implementation in every step that matters:

- its own edge order and its own encoding of the flags;
- its own 6-vertex representatives, by brute force over all 720 permutations of all 2^{15} labelled 4-graphs on six vertices, instead of nauty;
- an evaluator in C with 128-bit integers that enumerates every ordered injective θ with an exact labelled type test and every ordered disjoint pair (U_1, U_2) ; it does not use the automorphisms of σ ;
- a separate check that every flag decodes to a flag of its type, that the flags of a list are pairwise non-isomorphic and admissible, and that every list is complete.

For each of the five certificates it scanned all raw extensions of Table 3, found exactly the claimed minimum with the same group maxima as the first implementation, and evaluated the minimisers again with a pure-Python evaluator in exact fractions. Scanning the class lists of the first implementation gave the same result. Independently of the certificates, it recomputed the class counts by Burnside's lemma. With $Q = 0$ it recovered the minimum edge counts 7, 3 and 1 for $p = 5, 6, 7$, the Turán numbers $T(7, p, 4)$.

3.6 Finding a certificate

The search maximises b subject to $d(H) - \sum c_Q(H) \geq b$ for every admissible H and $Q \succeq 0$ for every block. Following Jeong et al. [17, Section 4], we restrict each matrix to nonnegative combinations of finitely many rank-one matrices, as in [1], and impose the graph constraints on a working set. The resulting linear program is solved with HiGHS [16]. Eigenvectors corresponding to negative eigenvalues of the dual moment matrices supply new rank-one terms; a scan over all admissible classes supplies violated graph constraints. After freezing the rank-one terms, we add graphs until a full scan finds no violated constraint. The search uses floating point; the proof uses only the rational matrices and exact minima described above.

4 Codegree-squared density and stability

We first prove the properties of Giraud systems needed for the exact value of $\sigma(K_5^{(4)})$. The sharp certificate then controls the proportion of seven-vertex subgraphs that are not Giraud systems. A local-to-global characterisation and a counting argument turn this control into an edit bound; the final step gives balanced parts and a quasirandom matrix.

4.1 Properties of Giraud systems

Lemma 10. *Every Giraud system is odd: every five vertices span 1, 3 or 5 edges. In particular it is a Turán $(n, 5, 4)$ -system. If $|A| = \alpha n$, $|B| = \beta n$, $s = \alpha\beta$ and f is the fraction of even minors among the pairs of pairs, then $d(\Gamma) = 1 - 4s + 2s^2 + 6s^2f + O(1/n)$, and $f \geq \frac{1}{2} - \frac{1}{2(\max(|A|, |B|)-1)}$ when both parts have at least two points.*

Proof. A five-set inside one part spans five edges. A five-set with four points in one part spans exactly one, the internal 4-set. A five-set $\{a_1, a_2, a_3, b, b'\}$ spans only the three cross sets $\{a_i, a_j, b, b'\}$; with $\rho_i = M_{a_i b} + M_{a_i b'}$ the three minors are $\rho_i + \rho_j$, whose sum is even, so one or three of them are even. The case of three points in B is symmetric.

For the density, $\alpha^4 + \beta^4 = 1 - 4s + 2s^2$ and $\binom{\alpha n}{2} \binom{\beta n}{2} = 6s^2 \binom{n}{4} + O(n^3)$. For f , assume $|B| \geq |A|$ (otherwise transpose M) and put $\varepsilon_{ab} = (-1)^{M_{ab}}$. Summing over ordered pairs $a \neq a'$ and $b \neq b'$,

$$\sum_{a \neq a'} \varepsilon_{ab} \varepsilon_{ab'} \varepsilon_{a'b} \varepsilon_{a'b'} = \sum_{a \neq a'} \left(\left(\sum_b \varepsilon_{ab} \varepsilon_{a'b} \right)^2 - |B| \right) \geq -|A|(|A| - 1)|B|.$$

There are $|A|(|A| - 1)|B|(|B| - 1)$ terms, so the average of $(-1)^{\text{minor}}$ is at least $-1/(|B| - 1)$, and $f \geq \frac{1}{2} - \frac{1}{2(|B| - 1)}$. $\square$

So $d(\Gamma) \geq 1 - 4s + 5s^2 - o(1) \geq 5/16 - o(1)$, with near-equality only for $s \approx 1/4$ and $f \approx 1/2$; these near-extremal members are Giraud's construction.

4.2 The exact value

For a 4-graph G on $n \geq 5$ vertices let

$$\gamma(G) = \mathbb{P}[T \cup \{x\} \in E(G) \text{ and } T \cup \{y\} \in E(G)],$$

where T is a uniform 3-subset of $V(G)$ and (x, y) is a uniform ordered pair of distinct vertices outside T .

Lemma 11. *Let G be a 4-graph on $n \geq 5$ vertices and H its complement in $K_n^{(4)}$. Then*

$$\frac{\text{co}_2(G)}{\binom{n}{3}(n-3)^2} = \gamma(G) - \frac{\gamma(G) - d(G)}{n-3} \quad \text{and} \quad \gamma(G) = 1 - \kappa(H),$$

where $\kappa(H) = 2d(H) - \gamma(H)$.

Proof. Write $\deg_G(T)^2 = \deg_G(T)(\deg_G(T) - 1) + \deg_G(T)$. Summed over T , the first terms count the triples (T, x, y) with $x \neq y$ outside T and $T \cup \{x\}, T \cup \{y\} \in E(G)$, so they add up to $\binom{n}{3}(n-3)(n-4)\gamma(G)$; the second terms add up to $4e(G) = \binom{n}{3}(n-3)d(G)$. Dividing by $\binom{n}{3}(n-3)^2$ gives the first identity. The second is inclusion-exclusion: each of $T \cup \{x\}$ and $T \cup \{y\}$ is an edge of H with probability $d(H)$. $\square$

Both d and γ are densities of configurations on at most five vertices, so κ has the property of Remark 8 for $N = 7$. On seven vertices $\kappa(H) = (24e(H) - P(H))/420$, where $P(H) = \sum_T \deg_H(T)(\deg_H(T) - 1)$ and $420 = \binom{7}{3} \cdot 4 \cdot 3$ is the number of triples (T, x, y) .

Proof of Theorem 4. Upper bound. A 4-graph G is $K_5^{(4)}$ -free iff its complement H is a Turán $(n, 5, 4)$ -system. Below we describe a plain certificate on seven vertices for $(p, \lambda) = (5, 1)$ with d replaced by κ , of value exactly $b = 33/64$. By Remark 8, $\kappa(H) \geq 33/64 - O(1/n)$ for every Turán $(n, 5, 4)$ -system H , so by Lemma 11, since $\gamma(G) - d(G) \geq -1$,

$$\frac{\text{co}_2(G)}{\binom{n}{3}(n-3)^2} \leq \gamma(G) + \frac{1}{n-3} = 1 - \kappa(H) + \frac{1}{n-3} \leq \frac{31}{64} + O(1/n)$$

for every $K_5^{(4)}$ -free G on n vertices, and $\sigma(K_5^{(4)}) \leq 31/64$.

Lower bound. This is the construction of [2, Section 8.2]; we give an explicit instance. Let $n = 2^{k+1}$, split the vertices into halves A and B , let M be the 0/1 form of the Sylvester-Hadamard matrix of order 2^k , so that any two distinct rows of M differ in exactly half of the columns and any two distinct columns in exactly half of the rows, and let G be the complement of the Giraud system $\Gamma(A, B, M)$ (Definition 3); G is $K_5^{(4)}$ -free by Lemma 10. The edges of G are the 4-sets with three points in one part and one in the other, and the 4-sets $\{a, a', b, b'\}$ with an odd minor. Up to $O(1/n)$ a uniform 3-set T lies inside one part with probability $1/4$, and then $T \cup \{x\}$ and $T \cup \{y\}$ are both edges iff x and y lie in the other part, with probability $1/4$. Otherwise, say $T = \{a, a', b\}$, the set $T \cup \{x\}$ is an edge iff $x \in A$, or $x \in B$ and $M_{ax} + M_{a'x} \neq M_{ab} + M_{a'b}$ modulo 2; the second condition holds for exactly half of the columns x , whatever b is. So both are edges with probability $\frac{1}{4} + 2 \cdot \frac{1}{4} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{4} = \frac{9}{16}$, and $\gamma(G) = \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{9}{16} + O(1/n) = \frac{31}{64} + O(1/n)$. By Lemma 11 the normalised $\text{co}_2(G)$ tends to $31/64$; since these G are $K_5^{(4)}$ -free, $\sigma(K_5^{(4)}) \geq 31/64$ (a subsequence suffices for the lim sup). $\square$

The certificate. Let ϕ be the law of the 7-vertex subgraph of Giraud’s construction on uniformly random vertices, with parts and matrix entries independent and uniform: $\mathbb{E}_\phi \kappa = 33/64$ exactly (the computation above, carried out in this model, has no error term), and ϕ is supported on ten classes, all of them Turán systems. Every plain certificate has $b \leq \mathbb{E}_\phi(\kappa - \sum c_Q) \leq 33/64$, because $\mathbb{E}_\phi c_Q = \sum_\omega \mathbb{P}(\omega) v_\omega^\top Q v_\omega \geq 0$, the sum over the root configurations ω (the parts of the s root vertices and the matrix entries among them), where v_ω is the law of the rooted flag given ω (flags on disjoint vertex sets are independent given ω). A certificate of value $33/64$ must therefore be tight on all ten classes and satisfy $v_\omega^\top Q v_\omega = 0$, or equivalently $(Q + Q^\top)v_\omega = 0$, for every ω of positive probability. Our matrices are symmetric, so this condition is $Qv_\omega = 0$.

To attain equality, rounding must preserve the conditions $Qv_\omega = 0$ and the value on the ten classes.

For every type τ the sharp certificate has the form

$$Q_\tau = W_\tau^\top W_\tau / 2^{72} + B_\tau^\top X_\tau B_\tau,$$

where the integer rows of W_τ are orthogonal to all vectors v_ω (the floating-point solution of a cutting-plane run restricted to such matrices, rounded with the pivot coordinates solved exactly), B_τ is an integer basis of the span of the flags occurring in Giraud’s construction intersected with the orthogonal complement of the v_ω , and $X_\tau = 2^{-20}I + \Delta_\tau$ with Δ_τ (entries below $4 \cdot 10^{-11}$) solved exactly so that $\kappa - \sum c_Q$ equals $33/64$ on each of the ten classes. The second term is present only for the three types of the $(5, 6)$ block with 1, 3 and 5 edges (dimensions 20, 5, 1), and each X_τ is strictly diagonally dominant, hence positive definite (also checked by an exact $LDL^\top$ factorisation); so $Q_\tau \succeq 0$. The entries of all Q_τ have common denominator $57 \cdot 2^{152}$. Both implementations evaluated $\kappa - \sum c_Q$ exactly on all 3 908 438 admissible classes and on all 86 952 880 raw extensions of Table 3: the minimum is exactly $33/64$, attained exactly on the ten classes of ϕ , and every other admissible 7-vertex graph exceeds it by at least $2.8 \cdot 10^{-5}$. Tightness on Giraud’s construction is not used for validity; it only explains why the bound is attained.

4.3 From the certificate to local structure

We prove Theorem 5. Except in Theorem 13 and its lemmas, H is a Turán $(n, 5, 4)$ -system, the complement of a $K_5^{(4)}$ -free G , and κ and γ are as in Lemma 11. On five vertices, $\kappa = g(e)$ with $g(e) = e(9 - e)/20$, because two distinct 4-subsets of a 5-set meet in a 3-set; so $\kappa(H) = \mathbb{E}_S g(e(H[S]))$ over uniform 5-sets S , and $g(1), \dots, g(5) = \frac{2}{5}, \frac{7}{10}, \frac{9}{10}, 1, 1$. Giraud systems (Definition 3) are *odd*: every five vertices span an odd number of edges (Lemma 10); and induced subgraphs of Giraud systems are Giraud systems. We call H *locally Giraud* if every 7-vertex induced subgraph is a Giraud system.

Finite facts. The following were checked exhaustively by two independent programs, which both enumerate Giraud systems from Definition 3; to extend by a vertex, the first solves the parity system for the new link and tests every 7-set, the second joins the admissible link patterns over all 6-subsets.

- (G7) There are 2 404 labelled Giraud systems on seven vertices, in 10 isomorphism classes; these are exactly the ten classes on which the certificate of Theorem 4 is tight. Only one class has more than one partition: the 4-graph whose edges are the complements of the seven lines of a Fano plane, with seven partitions of sizes $(3, 4)$.
- (G8) There are 32 981 labelled Giraud systems on eight vertices, in 22 classes. Only one class has more than one partition: the Steiner quadruple system $S(3, 4, 8)$ (the planes of $AG(3, 2)$), with seven partitions of sizes $(4, 4)$; every 5-set of it spans exactly one edge.

(Cn) For $n = 8, 9, 10$, every locally Giraud 4-graph on n vertices is a Giraud system. (For $n = 10$ the second program tested all extensions of all 604 426 labelled Giraud systems on nine vertices and found exactly the 15 636 107 labelled Giraud systems on ten vertices.)

Proposition 12. *Let $\mu_0 = 176165518826891630107/6280747422216628134215680 > 2.8048 \cdot 10^{-5}$, the least value of $\kappa - \sum c_Q - 33/64$ over the admissible 7-vertex graphs that are not Giraud systems, for the certificate of Theorem 4. For every Turán $(n, 5, 4)$ -system H on $n \geq 10$ vertices,*

$$\mu_0 q(H) \leq \xi(H), \quad \text{where } \xi(H) = \kappa(H) - \frac{33}{64} + \frac{68}{n-7}$$

and $q(H)$ is the fraction of 7-sets S such that $H[S]$ is not a Giraud system.

Proof. By Section 4.2, $\kappa - \sum c_Q \geq 33/64$ on every admissible 7-vertex graph, with equality exactly on ten classes, which by (G7) are the Giraud systems; so it is $\geq 33/64 + \mu_0$ on the others; μ_0 was computed by an exact scan of all 3 908 438 classes and agrees with the independent evaluator of Section 4.2. Since $\kappa(H) = \mathbb{E}_S \kappa(H[S])$ over 7-sets, $\kappa(H) \geq 33/64 + \mu_0 q(H) + \sum_{\text{blocks}} \mathbb{E}_S c_Q(H[S])$. For one block, $\mathbb{E}_S c_Q(H[S]) = \langle Q, \bar{M} \rangle$, where $\bar{M}$ is the joint law of the two flags at a uniform root θ with independent uniform U_1, U_2 conditioned to be disjoint. If p is the probability that U_1 and U_2 meet, then, with v_θ as in the proof of Proposition 7, $\mathbb{E}_\theta[v_\theta v_\theta^\top] = (1-p)\bar{M} + pX$ with $X \geq 0$ entrywise of total mass at most 1, so $\langle Q, \bar{M} \rangle \geq -\frac{p}{1-p} \max_a Q_{aa}$, since $Q \succeq 0$. The largest diagonal entries of the matrices Q , summed over the types of each block, are 61.2023, 0.9435, 1.4008 and 0.0003 for $(5, 6)$, $(4, 5)$, $(3, 5)$ and $(1, 4)$ (exact values from the certificate; the blocks $(2, 4)$ and $(3, 4)$ are zero), and $p/(1-p)$ is $1/(n-6)$, $1/(n-5)$, $(4n-18)/((n-5)(n-6)) \leq 4/(n-7)$ and at most $12/(n-7)$ for these blocks when $n \geq 10$. The total is at most $67.76/(n-7)$, so $\kappa(H) \geq 33/64 + \mu_0 q(H) - 68/(n-7)$. $\square$

The margin μ_0 is attained by an odd 7-vertex graph with eleven edges that is not a Giraud system; it is the bottleneck of all constants below.

4.4 Giraud systems are determined by their seven-vertex subgraphs

Theorem 13. *For $n \geq 7$, a 4-graph on n vertices is a Giraud system if and only if it is locally Giraud.*

For a 4-graph H and a vertex v , the *link* L_v is the 3-graph $\{R : R \cup \{v\} \in E(H)\}$ on $U = V(H) \setminus \{v\}$. Say that a 3-graph L on U has *form F with representation $A' \subseteq U$* , and $B = U \setminus A'$, if (F1) every triple inside A' is in L , (F2) no triple with exactly two points in A' is in L , (F3) no triple inside B is in L , and (F4) for $a \in A'$ and distinct $b_1, b_2, b_3 \in B$, an odd number of the triples $ab_i b_j$ are in L . By (F4) the graph $\{bb' : abb' \in L\}$ on B has one or three edges on every three vertices, so it is a disjoint union of at most two cliques.

Lemma 14. *If $H = \Gamma(A, B, M)$ and $v \in A$, then L_v has form F with representation $A \setminus \{v\}$.*

Proof. A triple inside $A \setminus \{v\}$ completes a 4-set inside A ; a triple with two points in A and one in B , or inside B , completes a $(3, 1)$ or a $(1, 3)$ set; and $\{v, a, b, b'\}$ is an edge iff $M_{vb} + M_{ab} = M_{vb'} + M_{ab'}$ modulo 2, an equivalence relation on B with at most two classes, which gives (F4). $\square$

Lemma 15. *If $A'_1 \neq A'_2$ are two representations of the same 3-graph L , then $|A'_1| \leq 3$ and $|A'_2| \leq 3$.*

Proof. By (F1)–(F3) every triple of L meets each A'_i in one or three points. Let $X = A'_1 \cap A'_2$, $Y = A'_1 \setminus A'_2$, $Z = A'_2 \setminus A'_1$; $Y \cup Z \neq \emptyset$. The triples inside A'_1 are in L , so none of them meets A'_2 in zero or two points; hence $|Y| \leq 2$, and $|X| \leq 1$ unless $Y = \emptyset$. Symmetrically $|Z| \leq 2$, and $|X| \leq 1$ unless $Z = \emptyset$. So $|X| \leq 1$ and $|A'_i| \leq 3$. $\square$

Lemma 16. *If $\Gamma(A, B, M) = \Gamma(A_2, B_2, M_2)$ and $|A| \geq 5$, then $\{A, B\} = \{A_2, B_2\}$. In particular a Giraud system on $n \geq 9$ vertices has a unique partition.*

Proof. Take $v \in A$ and let P_2 be the part of the second partition containing v . By Lemma 14, applied to both representations (with the parts swapped if $v \in B_2$), $A \setminus \{v\}$ and $P_2 \setminus \{v\}$ both represent L_v . The first has at least four points, so by Lemma 15 they are equal, and $P_2 = A$. $\square$

Proof of Theorem 13. Induced subgraphs of Giraud systems are Giraud systems, which gives one direction. For the other, $n = 7$ is the definition and $n = 8, 9$ are (C8) and (C9). Restrictions of a representation $\Gamma(A, B, M)$ to a subset W are representations of $H[W]$. Let H be locally Giraud on $n \geq 10$ vertices. We first show that there is a partition Π of V such that, for every vertex w , $H - w$ is a Giraud system whose unique partition π_w is the restriction of Π .

For $n \geq 11$ each $H - w$ is a Giraud system by induction, with a unique partition π_w (Lemma 16). For $w \neq w'$ the restrictions of π_w and $\pi_{w'}$ to $V \setminus \{w, w'\}$ ($n - 2 \geq 9$ vertices) are partitions of the same Giraud system, hence equal. So “ $x \sim y$ iff x and y lie in the same part of π_w for some $w \notin \{x, y\}$ ” does not depend on w ; any three points lie in a common $V \setminus \{w\}$, so $\sim$ is an equivalence relation with at most two classes, and these form Π .

For $n = 10$ each $H - w$ is a Giraud system by (C9), with a unique partition π_w of sizes $(a, 9 - a)$. *Case 1: some $\pi_{w_0} = \{P, Q\}$ has $|P| \geq 6$.* For $x \neq w_0$, the restriction $\{P \setminus \{x\}, Q \setminus \{x\}\}$ of π_{w_0} to $V \setminus \{x, w_0\}$ has a part of size at least five, so it is the only partition there, and π_x is obtained from it by adding w_0 to one side, say to the side of P or of Q . If w_0 joins P in π_x , then for $x' \neq x, w_0$ the restriction of π_x to $V \setminus \{x, x'\}$ has the part $(P \setminus \{x, x'\}) \cup \{w_0\}$ of size at least five, so it is the unique partition there, and w_0 joins P in $\pi_{x'}$ too. So the side of w_0 is the same for all x , and $\Pi = \{P \cup \{w_0\}, Q\}$ or $\{P, Q \cup \{w_0\}\}$. *Case 2: every π_w has sizes $(5, 4)$.* Fix w_0 , $\pi_{w_0} = \{P, Q\}$ with $|P| = 5$, and $R = Q \cup \{w_0\}$. For $y \in Q$, the restriction $\{P, Q \setminus \{y\}\}$ to $V \setminus \{w_0, y\}$ is the unique partition there, so π_y is $\{P, R \setminus \{y\}\}$ or $\{P \cup \{w_0\}, Q \setminus \{y\}\}$, and the latter has sizes $(6, 3)$; so $\pi_y = \{P, R \setminus \{y\}\}$ for every $y \in R$. Now let $x \in P$. For $y \in R$ the 8-set $V \setminus \{x, y\}$ has the partition $\{P \setminus \{x\}, R \setminus \{y\}\}$; call y good if this is its only partition. If y is good, π_x is $\{P \setminus \{x\}, R\}$ or $\{(P \setminus \{x\}) \cup \{y\}, R \setminus \{y\}\}$, so two good y force $\pi_x = \{P \setminus \{x\}, R\}$. Otherwise at least four $y \in R$ are bad, and by (G8) every 5-subset of $V \setminus \{x, y\}$ spans exactly one edge for each bad y . The 5-part P_x of π_x spans five edges, so it contains every bad y . If all five are bad, $P_x = R$. If exactly four are bad, forming Y , then $P_x = Y \cup \{z\}$; if $z \in R$ then $P_x = R$, and $z \in P \setminus \{x\}$ is impossible: for the good y_5 the restriction $\{Y \cup \{z\}, P \setminus \{x, z\}\}$ of π_x to $V \setminus \{x, y_5\}$ has a 5-part, so it is the unique partition there, but the restriction of π_{y_5} is $\{P \setminus \{x\}, Y\}$. So $\pi_x = \{P \setminus \{x\}, R\}$ in all cases, and $\Pi = \{P, R\}$.

Gluing. Let $\Pi = \{A, B\}$. If $B = \emptyset$, every $H - w$ is complete, and so is H . Otherwise fix $a_0 \in A$, $b_0 \in B$, and put $M_{ab} = 1 - [\{a_0, a, b_0, b\} \in E(H)]$ for $a \neq a_0$, $b \neq b_0$, and $M = 0$ on row a_0 and column b_0 . For a 4-set T choose $w \notin T \cup \{a_0, b_0\}$. Then $H - w = \Gamma(A \setminus \{w\}, B \setminus \{w\}, M_w)$, and after switching M_w to vanish on row a_0 and column b_0 its entries are those of M (they are read off the same edges $\{a_0, a, b_0, b\}$). So $T \in E(H)$ iff $T \in E(\Gamma(A, B, M))$, and $H = \Gamma(A, B, M)$. $\square$

The case $n = 10$, the only one used below, was also checked exhaustively by both programs (C10). On at least seven vertices, the co-Fano class and $S(3, 4, 8)$ are the only Giraud systems with more than one partition (by (G7), (G8) and Lemma 16), which is why $n = 8, 9$ are done

by computer and $n = 10$ needs Case 2. Chen, Hou, Liu, Zhang and Zhao [5, Lemma 3.3] prove a six-vertex local-to-global lemma of the same kind for complete odd-bipartite 4-graphs.

4.5 Robust stability

Theorem 17. *For every Turán $(n, 5, 4)$ -system H on $n \geq 10$ vertices there is a Giraud system Γ on $V(H)$ with*

$$|E(H) \triangle E(\Gamma)| \leq \left(C_0 \xi(H) + \frac{24}{n} \right) \binom{n}{4}, \quad C_0 = \frac{21894}{\mu_0} < 7.81 \cdot 10^8.$$

Consequently every $K_5^{(4)}$ -free G on $n \geq 10$ vertices with $\text{co}_2(G) \geq (31/64 - \varepsilon) \binom{n}{3} (n-3)^2$ is within $(C_0(\varepsilon + \frac{1}{n-3} + \frac{68}{n-7}) + \frac{24}{n}) \binom{n}{4}$ changed 4-sets of the complement of a Giraud system.

Proof. A frame is an ordered 6-tuple $\Phi = (a_0, a_1, a_2, a_3, a_4, b_0)$ of distinct vertices such that $\{a_0, \dots, a_4\}$ spans five edges and $\{a_0, a_1, a_2, b_0\}$ is not an edge. It defines a Giraud system $\Gamma_\Phi = \Gamma(A_\Phi, B_\Phi, M_\Phi)$: A_Φ consists of a_0, a_1, a_2 and the vertices x with $\{a_0, a_1, a_2, x\} \in E(H)$, B_Φ of the others (so $a_3, a_4 \in A_\Phi$ and $b_0 \in B_\Phi$), M_Φ vanishes on row a_0 and column b_0 , and $M_\Phi(a, b) = 1 - [\{a_0, a, b_0, b\} \in E(H)]$ otherwise.

Claim: if T is a 4-set disjoint from Φ and $H[\Phi \cup T]$ is a Giraud system, then $T \in E(H)$ iff $T \in E(\Gamma_\Phi)$. Write $H[\Phi \cup T] = \Gamma(A', B', M')$. A 5-set spans five edges only inside a part, so $\{a_0, \dots, a_4\} \subseteq A'$, say, and $b_0 \in B'$ because $\{a_0, a_1, a_2, b_0\}$ is not an edge. For $x \in T$, $\{a_0, a_1, a_2, x\}$ is an edge iff $x \in A'$, so $A' = A_\Phi \cap (\Phi \cup T)$. After switching M' to vanish on row a_0 and column b_0 , the minor of $\{a_0, a, b_0, b\}$ is $M'(a, b)$, so $M' = M_\Phi$ on $\Phi \cup T$, and $H[\Phi \cup T] = \Gamma_\Phi[\Phi \cup T]$.

Counting. Call a 10-set bad if it does not induce a Giraud system; by Theorem 13 (at $n = 10$) it then contains a 7-set that does not induce one, so there are at most $\binom{n}{7} \binom{n-7}{3} q(H) = 120 q(H) \binom{n}{10}$ bad 10-sets. A pair of a frame Φ and a 4-set T disjoint from it with $[T \in E(H)] \neq [T \in E(\Gamma_\Phi)]$ makes $\Phi \cup T$ bad, and a 10-set arises from at most $(10)_6$ such pairs. Let ϱ be the number of frames divided by $(n)_6$. Averaging over frames, some frame Φ has at most $120 q(H) \binom{n}{10} (10)_6 / (\varrho (n)_6) = (120 q(H) / \varrho) \binom{n-6}{4}$ such 4-sets T , and at most $6 \binom{n-1}{3} = \frac{24}{n} \binom{n}{4}$ further 4-sets meet Φ . So $|E(H) \triangle E(\Gamma_\Phi)| \leq (120 q(H) / \varrho + 24/n) \binom{n}{4}$.

Frame density. Frames live on six vertices, so ϱ is the average of the frame densities of the 10-vertex induced subgraphs. A Giraud system on ten vertices that is not complete has parts of sizes $a \geq 10 - a \geq 1$ and at least $(a)_5 (10 - a) + (10 - a)_5 a \geq 1200 = (10)_6 / 126$ frames. Complete 10-sets have $\kappa = 1$ and every admissible graph has $\kappa \geq 2/5$, so the fraction w of complete 10-sets satisfies $\kappa(H) \geq w + \frac{2}{5}(1 - w)$. Hence $\varrho \geq (1 - \frac{5}{3}(\kappa(H) - \frac{2}{5}) - 120 q(H)) / 126$.

Constants. If $\xi(H) \leq \mu_0 / 1200$, then $120 q(H) \leq 1/10$ by Proposition 12 and $\kappa(H) \leq 33/64 + 1/100$, so $\varrho \geq \frac{221}{320} \cdot \frac{1}{126}$ and $120 q(H) / \varrho \leq 21893.3 q(H) \leq C_0 \xi(H)$. If $\xi(H) > \mu_0 / 1200$, the bound exceeds $\binom{n}{4}$. For the second statement, Lemma 11 and $d - \gamma \leq 1$ give $\kappa(H) - 33/64 = 31/64 - \gamma(G) \leq 31/64 - \text{co}_2(G) / (\binom{n}{3} (n-3)^2) + 1/(n-3)$. $\square$

Proof of Theorem 5. By Theorem 17 it suffices that $C_0 \varepsilon \leq \delta/2$ and $(69C_0 + 24)/(n-7) \leq \delta/2$. For the structure, let $\Gamma = \Gamma(A, B, M)$ be the Giraud system of Theorem 17, $s = |A||B|/n^2$ and f its fraction of even minors. Counting the 5-sets by type, as in Lemma 10, gives

$$\kappa(\Gamma) = 1 - 3s + 3s^2 + \frac{5}{2}(3f - 1)s^2 + O(1/n), \quad \kappa(\Gamma) - \frac{33}{64} \geq \frac{7}{8}(\frac{1}{4} - s) + \frac{15}{2}s^2(f - \frac{1}{2}) - O(1/n),$$

and $f \geq \frac{1}{2} - O(1/n)$ when both parts have at least two points (Lemma 10); otherwise $s = O(1/n)$. Changing one 4-set changes κ by at most $2/\binom{n}{4}$, because $|g(e) - g(e \pm 1)| \leq 2/5$. So $\kappa(\Gamma) - 33/64 = O(\varepsilon + 1/n)$, hence $\frac{1}{4} - s = (\frac{1}{2} - |A|/n)^2 = O(\varepsilon + 1/n)$ and $f = \frac{1}{2} + O(\varepsilon + 1/n)$. $\square$

The condition $f = \frac{1}{2} + o(1)$ is the 4-cycle condition: the ± 1 matrix $((-1)^{M_{ab}})$ has $\sum_{a \neq a'} \langle r_a, r_{a'} \rangle^2 = o(|A|^2|B|^2)$ for its rows r_a , the 4-cycle form of quasirandomness [7], invariant under switching. So near-extremal $K_5^{(4)}$ -free graphs are close to the complement of Giraud's construction: balanced parts and a quasirandom matrix. In particular $d(G) = 11/16 + O(\varepsilon + 1/n)$, by the density formula of Lemma 10. A uniformly random matrix gives some balanced Giraud system with $f \leq \frac{1}{2}$, so every $K_5^{(4)}$ -free G with the largest $\text{co}_2(G)$ on n vertices has $\varepsilon = O(1/n)$, and by Theorem 17 it is within $O(n^3)$ changed 4-sets of the complement of a Giraud system.

Remark 18. The constants are explicit but not optimised, and they are weak because μ_0 is small: Proposition 12 says something only for $n > 2.4 \cdot 10^6$, and Theorem 17 only for $n > 5.38 \cdot 10^{10}$ and $\varepsilon < 1.3 \cdot 10^{-9}$. The qualitative statement, Theorem 5, holds with ε linear in δ , and its proof uses no removal lemma. A certificate with a larger margin outside Giraud's ten classes would improve all constants linearly.

Remark 19. The exact structure for large n remains open: we expect that every $K_5^{(4)}$ -free 4-graph with the largest codegree-squared sum is the complement of a Giraud system.

Theorems 5 and 17 concern κ , not the edge density: they do not imply Hypothesis 40. By the Cauchy–Schwarz inequality $\gamma(H) \geq d(H)^2 - O(1/n)$, so Theorem 4 gives only $t(5, 4) \geq 1 - \sqrt{31}/8 > 0.30402$, weaker than Theorem 2(a).

5 Limits of the six- and seven-vertex method

A *dual point on N vertices* is a probability vector $y = (y_H)$ on the admissible 4-graphs on N vertices (up to isomorphism) such that for every type σ of every pair (s, m) with $2m - s \leq N$ the moment matrix $Y_\sigma = \sum_H y_H M_\sigma(H)$, indexed by all admissible σ -flags on m vertices, is positive semidefinite.

Lemma 20. *If y is a dual point on N vertices, then every plain certificate on N vertices has value $b \leq \sum_H y_H d(H)$.*

Proof. $b \leq \sum_H y_H (d(H) - \sum c_Q(H)) = \sum_H y_H d(H) - \sum \langle Q', Y_\sigma \rangle$, where Q' is the principal submatrix of Q on the admissible flags of its list, extended by zero to all admissible flags (inadmissible flags never occur in an admissible graph). Each $\langle Q', Y_\sigma \rangle$ is nonnegative because Y_σ is symmetric (the two flags of a configuration play symmetric roles), so only the symmetric part of Q' matters, and both are positive semidefinite. $\square$

5.1 The six-vertex barrier

Recall Definition 6: “plain” means that only sums of squares of flag densities are used, with no stationarity or other additional constraint.

Theorem 21. *No plain flag-algebra certificate on six vertices proves $t(5, 4) > 1/4$, $t(6, 4) > 1/10$, $t(7, 4) > 0$ or $t_2(5, 4) > 1/2$.*

In Turán form the six-vertex method cannot go below $3/4$, $9/10$ and $1/2$ for $K_5^{(4)}$, $K_6^{(4)}$ and $K_7^{(4)}$, which are de Caen's bounds and the general $(k-2)/r$ bound, and it gives nothing for $K_7^{(4)}$. The values $1/4$, $1/10$ and $1/2$ are attained by six-vertex certificates, checked exactly by the first implementation and by an independent checker (Remark 23), so they are the exact six-vertex optima.

Proof of Theorem 21. At $N = 6$ the pairs are $(0, 1)$, $(0, 2)$, $(0, 3)$, $(1, 2)$, $(1, 3)$, $(2, 3)$, $(2, 4)$, $(3, 4)$ and $(4, 5)$, with two types for $(4, 5)$; pairs with $m = s$ have one flag per type, and their

1×1 moment matrices are nonnegative. The dual points of Table 4 have all ten moment matrices positive semidefinite and give $\sum_H y_H d(H) = 1/4, 1/10, 0$ and $1/2$. Lemma 20 completes the proof. $\square$

The positive semidefiniteness of the moment matrices was verified in exact arithmetic by two implementations, from the definitions (own flag enumeration, own densities, exact LDL^T test). For $t(5, 4)$ and $t(6, 4)$ all ten matrices have rank one; for $t(7, 4)$ nine have rank one and one is zero.

problem	support (pair graphs on $\{0, \dots, 5\}$)	weights	$\sum y_H d(H)$
$t(5, 4)$	perfect matching; star $K_{1,5}$; K_6	$45/64, 9/32, 1/64$	$1/4$
$t(6, 4)$	one pair; perfect matching; star $K_{1,5}$; K_6	$81/100, 27/200, 27/500, 1/1000$	$1/10$
$t(7, 4)$	the empty graph	1	0
$t_2(5, 4)$	$K_{1,1,4}$; C_6 ; two disjoint triangles; K_6	$5/12, 5/12, 5/36, 1/36$	$1/2$

Table 4: Dual points on six vertices. A 4-graph on six vertices is described by the graph of the complementary pairs of its edges; for instance “star $K_{1,5}$ ” is $K_5^{(4)}$ plus an isolated vertex.

Remark 22. On six vertices a 4-graph is a graph in disguise: the map $T \mapsto \{0, \dots, 5\} \setminus T$ sends 4-sets to pairs, and the five-vertex subsets correspond to the vertices. A 4-graph on six vertices is 5-admissible iff its pair graph has no isolated vertex, and 6-admissible iff it has an edge. This is why six vertices see so little of the problem. On seven vertices the same map sends 4-graphs to 3-graphs.

Remark 23. Six-vertex certificates of value exactly $1/4, 1/10$ and $1/2$ exist; ours were checked exactly by the first implementation on all 2^{15} labelled 4-graphs on six vertices, and again by an independent checker of the data package written from the definitions (minimum over all admissible classes, attained on 3, 4 and 4 classes, the supports of the dual points). The lower bounds themselves are de Caen’s [8] and the general bound $\pi(H_k^r) \leq (k-2)/r$. So the plain six-vertex optimum is exactly de Caen’s bound for $K_5^{(4)}$ and $K_6^{(4)}$ and the $(k-2)/r$ bound for K_5^{4-} , and every improvement in Theorem 2(a)–(d) comes from seven-vertex flags. The statement is about the plain method only: Theorem 21 says nothing about six-vertex computations with additional constraints.

5.2 The seven-vertex limits

The same weak-duality argument shows that, for $K_5^{(4)}$, the seven-vertex method is essentially exhausted.

Theorem 24. *No plain flag-algebra certificate on seven vertices proves*

$$t(5, 4) > V = \frac{1336237682928914994292138923}{7 \cdot 2^{89}} = 0.308401201194\dots,$$

that is, none proves $\pi(K_5^{(4)}) < 0.691598798805$. With Theorem 2(a), the best plain seven-vertex bound lies in $[0.308400990997, 0.308401201195]$, an interval of width less than $2.2 \cdot 10^{-7}$. The same holds for the problem restricted to odd 4-graphs.

Proof. At $N = 7$ the pairs (s, m) with $m > s$ and $2m - s \leq 7$ are $(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3), (1, 4), (2, 4), (3, 4), (3, 5), (4, 5)$ and $(5, 6)$; the pairs with $m = s$ are also allowed. When $m \leq 3$ or $m = s$ there is one flag per type, so the moment matrix is a nonnegative 1×1 matrix. For the other six pairs we use the dual point y with weights $y_H = \nu_H / 2^{110}$ (ν_H positive integers summing to 2^{110}) on 131 admissible 7-vertex 4-graphs, all of them odd. Every

moment matrix Y_σ , for every labelled type σ of the six pairs, is positive semidefinite, and $\sum_H y_H d(H) = V$. Lemma 20 (which holds verbatim at $N = 7$) gives the first claim. Because every support graph is odd, the same y is a dual point for the odd-restricted problem. $\square$

The positive semidefiniteness was verified exactly by two independent implementations from the definitions. Rows and columns with zero diagonal vanish identically and are removed; the remaining parts of the $(5, 6)$ matrices have dimensions 64, 58 and 64 (types with 1, 3 and 5 edges; types with 0, 2 or 4 edges do not occur in the support), and each is positive definite on its occurring flags. Moment matrices of blocks with smaller flags are obtained from these by summing flags, so the argument also covers certificates that mix flag sizes.

The theorem does not cover stationarity constraints or more than seven vertices.

The same construction gives exact dual points for the other certificates of Theorem 2 (Table 5), again verified exactly by an independent implementation.

Without the $(5, 6)$ block, that is, for certificates whose types of size five enter only through one-flag terms, no plain seven-vertex certificate proves $t(5, 4) > 0.281644555065$; so the $(5, 6)$ block is worth at least 0.0267 in $t(5, 4)$.

problem	certified b (Theorem 2)	exact dual point V (rounded up)	$V - b$
$t(5, 4)$	0.308400990997	0.308401201195	$2.1 \cdot 10^{-7}$
$t(6, 4)$	0.139827122482	0.139837689411	$1.1 \cdot 10^{-5}$
$t(7, 4)$	0.064762076605	0.064764464150	$2.4 \cdot 10^{-6}$
$t_2(5, 4)$	0.552992751369	0.553946861054	$9.5 \cdot 10^{-4}$
$t_2(6, 4)$	0.219452823117	0.220748423194	$1.3 \cdot 10^{-3}$
$t(5, 4)$, no $(5, 6)$ block	—	0.281644555065	—

Table 5: What the plain seven-vertex method can give: no plain seven-vertex certificate proves more than V . The supports of the dual points are odd only for $t(5, 4)$ (Theorem 24).

For comparison, at $r = 3$ the plain method on $r + 2 = 5$ vertices already beats the bound of Chung and Lu for $K_4^{(3)}$: it gives $t(4, 3) > 0.426923$, that is $\pi(K_4^{(3)}) < 0.573077$, against $\pi(K_4^{(3)}) \leq (3 + \sqrt{17})/12 < 0.593593$ [6]. The first implementation found this value; an independent script written from the definitions (in the data package [25]) gives the exact certificate $t(4, 3) \geq 60084175574225/2^{47} > 0.4269237$ over all 768 labelled admissible 3-graphs on five vertices. And for $r = 5$ the seven-vertex computation of Theorem 26 beats it for $K_6^{(5)}$; among $r = 3, 4, 5$, only for $r = 4$ does the $(r + 2)$ -vertex method stop at de Caen's bound.

5.3 Parity

Parity in the seven-vertex certificates. Recall that a 4-graph is odd if every five vertices span an odd number of edges (Section 4.3). Giraud's construction is odd (Lemma 10). The dual point of Theorem 24 is supported on odd graphs. The certificate itself does not force parity: its minimum b is attained at a graph with fourteen even five-sets, while every odd 7-vertex 4-graph (there are 423 classes) has $d - \sum c_Q \geq b + 376193/28862180229120$. Restricted to odd 4-graphs, the certificate of Theorem 2(a) itself is worth $b + 376193/28862180229120 > 0.3084010040$, by the margin just stated, and no plain certificate is worth more than V (the dual point of Theorem 24 is supported on odd graphs); at seven vertices parity is not the missing information.

Remark 25. A 4-graph A on V is odd iff $A = \{T : T \text{ contains an even number of edges of } J\}$ for some 3-graph J on V . Indeed, a 4-set is a 3-simplex, and A is odd iff the indicator of the complement of A is a 3-cocycle of the full simplex on V over $\mathbb{F}_2$; every such cocycle is the coboundary of a 2-cochain, the indicator of a 3-graph J . So the question whether every odd 4-graph has density at least $5/16 - o(1)$, a weak form of Sidorenko's conjecture, is a question about 3-graphs.

5.4 Giraud's construction and the certificate

Take Giraud's construction with equal parts and independent uniform matrix entries. Its random 7-vertex subgraph has an explicit law: each vertex lies in a uniformly random part, and the entries of the matrix on the sampled rows and columns are independent uniform bits. The law is supported on 2404 labelled graphs in 10 isomorphism classes, all odd. For the certificate of Theorem 2(a) we computed exactly $\mathbb{E}d = 5/16$, $\mathbb{E}(d - \sum c_Q) = 173614396936021/2^{49} > 0.3084011214$ and $\mathbb{E}\sum c_Q = 2307463508139/2^{49} > 0.0040988785$, and every support graph satisfies $b + 7.09 \cdot 10^{-8} \leq d - \sum c_Q \leq b + 4.53 \cdot 10^{-7}$. A Giraud system of density $5/16 + o(1)$ has parts of size $n/2 + o(n)$ and a fraction $1/2 + o(1)$ of even minors (Lemma 10). Every Giraud system on seven vertices lies in the support of this law ((G7) in Section 4.3), so every 7-vertex subgraph of every Giraud system is tight for the certificate up to $4.6 \cdot 10^{-7}$, and the remaining gap $5/16 - b$ exceeds the sum-of-squares term of the law by about $1.3 \cdot 10^{-7}$. Closing the gap needs larger flags or additional structure, such as stability. For the codegree-squared objective the same law is tight exactly (Section 4.2).

6 Further bounds

The same certificate method gives bounds for 5-graphs, threshold families of 4-graphs and further codegree-squared densities. We give the statements, comparisons and verification details together.

6.1 Five-graphs

Theorem 26. $\pi(K_6^{(5)}) < 0.767401$, $\pi(H_4^5) < 0.389709$ and $\pi(K_7^{(5)}) < 0.927918$. The exact rational bounds are given in Appendix A.

Definition 6 and Proposition 7 hold verbatim for r -graphs, with r -sets in place of 4-sets, and with the same proof. Theorem 26 uses them for 5-graphs on seven vertices, with all blocks satisfying $2m - s \leq 7$. A 5-graph on seven vertices corresponds to a graph on seven vertices, so there are only 1044 classes, and the semidefinite program was solved directly. The matrices Q were rounded through their eigendecomposition, eigenvalues and eigenvector entries to multiples of 2^{-40} , which keeps them positive semidefinite. The certificate files store the rational matrices Q themselves, so a verifier must test positive semidefiniteness; both implementations did so exactly, by an LDL^T factorisation over the rationals, block by block. Both implementations evaluated the certificates on all 2^{21} labelled 5-graphs on seven vertices. The analogous computations for H_3^5 and H_5^5 give only the general bounds $1/5$ and $3/5$, up to rounding.

6.2 Threshold families

The same method applies to every threshold family. Let $F_{p,\lambda}$ be the family of p -vertex 4-graphs with at least $\binom{p}{4} - \lambda + 1$ edges; a 4-graph is $F_{p,\lambda}$ -free iff its complement spans at least λ edges in every p -set, so $\pi(F_{p,\lambda}) = 1 - t_\lambda(p, 4)$. The cases $(5, 1)$, $(5, 2)$, $(6, 1)$, $(6, 2)$ and $(7, 1)$ are $K_5^{(4)}$, K_5^{4-} , $K_6^{(4)}$, K_6^{4-} and $K_7^{(4)}$.

Theorem 27. The lower bounds on $t_\lambda(p, 4)$ in Table 6 hold; each is a plain seven-vertex certificate with exact rational entries, and the case $(5, 3)$ reproduces the known value $\pi(H_3^4) = 1/4$ [15] up to $1.7 \cdot 10^{-8}$.

The upper bounds come from inclusions and simple constructions. If every 5-set spans at least μ edges, every 6-set spans at least 3μ (each 4-set lies in two of its six 5-subsets), and if

(p, λ)	minimal members of $F_{p,\lambda}$	$t_\lambda(p, 4) >$	previous	construction	gap removed
(5, 3)	H_3^4	0.749999983648	3/4 (exact)	3/4	validation
(6, 3)	$K_6^{(4)}$ minus 2 edges	0.290180099259	0.219452	5/16	76.0%
(6, 4)	$K_6^{(4)}$ minus 3 edges	0.356110768826	0.291249	0.410755	54.3%
(6, 5)	$K_6^{(4)}$ minus 4 edges	0.421440536123	0.358011	1/2	44.7%
(6, 6)	$K_6^{(4)}$ minus 5 edges	0.497081561103	0.409140	4/7	54.2%
(6, 9)	6-vertex, 7 edges	0.705293990330	0.609234	3/4	68.2%
(6, 11)	6-vertex, 5 edges	0.840815156178	0.752169	19/21	58.1%
(7, 2)	K_7^{4-}	0.101742874380	0.069349	0.154102	38.2%
(7, 3)	$K_7^{(4)}$ minus 2 edges	0.131341381781	0.093111	0.154102	62.7%
(7, 4)	$K_7^{(4)}$ minus 3 edges	0.161680760732	0.123665	7/27	28.0%

Table 6: Threshold families at $N = 7$ (lower bounds floored; $\pi(F_{p,\lambda}) = 1 - t_\lambda(p, 4)$). “Previous” is the best lower bound available without these runs (repeated averaging from small exact values, and our bounds for K_6^{4-} and $K_7^{(4)}$); “construction” is the best upper bound on $t_\lambda(p, 4)$ we know (see the text).

every 6-set spans at least μ , every 7-set spans at least $\lceil 7\mu/3 \rceil$. So Giraud’s construction bounds $t_3(6, 4)$ by 5/16, Sidorenko’s constructions [34, 36] bound $t_2(7, 4)$, $t_3(7, 4)$ by 0.154102 and $t_6(6, 4)$ by 4/7, and the complement of the tournament 4-graph of Gunderson and Semeraro [15], in which every five vertices span 0 or 2 edges, bounds $t_9(6, 4)$ by 3/4. The remaining ones are elementary: the even bipartite 4-graph (the 4-sets meeting a fixed half of the vertices in 0, 2 or 4 points) has density 1/2 and every six vertices span at least five edges; three parts of relative sizes 0.18614, 0.40693, 0.40693 with all 4-sets inside a part and those of patterns (3, 1, 0), (3, 0, 1), (1, 3, 0), (1, 0, 3), (2, 1, 1), (0, 2, 2) give $t_4(6, 4) < 0.410755$; three equal parts with the 4-sets inside a part or with two points in each of two parts give $t_4(7, 4) \leq 7/27$; and the complement of the iterated complete 4-partite 4-graph gives $t_{11}(6, 4) \leq 19/21$. Apart from our own K_6^{4-} result we found nothing published on these families for $p = 6, 7$ and $\lambda \geq 2$, which are the dense regime of the Brown–Erdős–Sós problem. Individually the bounds are minor; their value is as a systematic table.

The certificates have $Q = A^\top A/M^2$ with $M = 2^{19}, \dots, 2^{22}$ and the blocks of Section 3.2; their exact values are in Appendix A. The independent implementation checked all ten by both class-list scans and raw-extension scans, with the same exact values.

6.3 Further codegree-squared densities

For a 3-graph F the codegree-squared density $\sigma(F)$ is defined as in Section 2, with pairs in place of triples and the normalisation $\binom{n}{2}(n-2)^2$ [2, eq. (1) of arXiv v3]. The proof of Lemma 11 gives, for a 3-graph G on $n \geq 4$ vertices, $\text{co}_2(G)/(\binom{n}{2}(n-2)^2) = \gamma(G) - (\gamma(G) - d(G))/(n-2)$, where $\gamma(G)$ is now the probability that $P \cup \{x\}$ and $P \cup \{y\}$ are both edges for a uniform pair P and a uniform ordered pair (x, y) of distinct vertices outside P . Let J_t be the 3-graph on $t+1$ vertices consisting of a vertex u and all triples $\{u, a, b\}$ with a, b among the other t vertices; a 3-graph is J_t -free iff no vertex link contains K_t . Let $K_5^<$ be the 3-graph on five vertices with eight edges whose two missing edges meet in exactly two vertices [2, Section 3.20 of arXiv v3], let K_5^- and K_5^{3-} (K_5^- in [2]) be obtained from $K_5^{(3)}$ by deleting two edges that meet in exactly one vertex, and one edge, respectively [2, Sections 3.19 and 3.21 of arXiv v3] (as $K_5^- \subset K_5^{3-}$, $\sigma(K_5^-) \leq \sigma(K_5^{3-})$), and let C_5 be the tight 5-cycle, with edges $\{i, i+1, i+2\}$ modulo 5 [2, Table 2 of arXiv v3]. Balogh, Clemen and Lidický conjecture $\sigma(J_4) = 0.28$ [2, Conjecture 3.4 of arXiv v3], on the basis of the iterated blow-up of the complement of the Fano plane of Bollobás, Leader and Malvenuto [4], and $\sigma(K_5^<) = 1/3$ [2, Conjecture 3.7 of

arXiv v3], the value of Turán’s construction for the tetrahedron $K_4^{(3)} \subset K_5^<$.

The constructions for J_4 and J_5 .

Proposition 28. $\sigma(J_4) \geq 16/57 = 0.2807017\dots$, $\sigma(J_5) \geq 36/91 = 0.3956043\dots$ and $\pi(J_5) \geq 3/5$.

Proof. For J_4 take the iterated blow-up of the complement of the Fano plane (iterated inside every part; density $1/2$); for J_5 the iterated blow-up of the complement of the affine plane $AG(2, 3)$. In both, every vertex link is properly $(t - 1)$ -colourable, so it contains no K_t : write vertices as words over the points of the plane and colour a vertex y of the link of x by the line through x_i and y_i , where i is the first coordinate in which x and y differ. (Iterated blow-ups of J_t -free non-degenerate 3-graphs are J_t -free in general [11].) For the first construction on $n = 7^L$ vertices, with $m = 7^{L-1}$, the codegree-squared sum satisfies $c_L = 7c_{L-1} + 336m^4$, since each of the 21 pairs of distinct parts contributes m^2 vertex pairs of codegree $4m$ (four of the five other points of the Fano plane complete the pair to a non-line), and pairs inside a part have all their codegree inside the part; so $\text{co}_2 = \frac{8}{57}(n^4 - n)$ and the normalised value tends to $16/57$; the second gives $36/91$ and edge density $3/5$. The best previous lower bounds we found are $\pi(J_5) \geq 1/2$ and $\sigma(J_5) \geq 0.28$ [2, Section 3.16 of arXiv v3], against the upper bounds 0.64474184 and 0.4427457 of [2]. $\square$

Remark 29. The value 0.28 in [2, Conjecture 3.4 of arXiv v3] comes from an arithmetic slip, identical in all three arXiv versions: the sum $\sum_i (1/7^4)^i$ there should be $\sum_i (1/343)^i$ for the construction described (their sum is that of the variant iterated inside a single part). So the natural corrected conjecture is $\sigma(J_4) = 16/57$. The same slip affects the bound $\sigma(J_5) \geq 0.28$ of [2]. The construction for J_5 in Proposition 28 is the next case of the same construction; we found it nowhere in the literature, but it is the obvious analogue and may be folklore.

Certified upper bounds.

Theorem 30. *The upper bounds in the last column of Table 7 hold; their exact values are the three below and those in Appendix A. In particular*

$$\begin{aligned}\sigma(J_4) &\leq \frac{9061620987350745659}{32281802128991715328} < 0.280703690307, \\ \sigma(K_5^<) &\leq \frac{1375676321453}{4123168604160} < 0.333646, \\ \sigma(K_5^{4-}) &\leq \frac{6281420455337}{30786325577728} < 0.204033.\end{aligned}$$

Proof. Definition 6, Proposition 7 and Remark 8 hold verbatim for 3-graphs and for F -free graphs in place of (p, λ) -admissible ones: the proof uses only that the class of graphs is closed under induced subgraphs. For the 3-graphs the blocks are the nine pairs (s, m) with $s < m$, $m \geq 3$ and $2m - s \leq 7$, and the objective is $1 - \gamma$, a density on four vertices; a certificate of value b gives $\gamma(G) \leq 1 - b + O(1/n)$ for every F -free G , hence $\sigma(F) \leq 1 - b$ by the identity above. For the 4-graphs, a 4-graph is K_5^{4-} -, K_6^{4-} -, $K_6^{(4)}$ - or $K_7^{(4)}$ -free iff its complement is $(5, 2)$ -, $(6, 2)$ -, $(6, 1)$ - or $(7, 1)$ -admissible, and the certificate is one for that (p, λ) with d replaced by κ , as in the proof of Theorem 4. Each certificate has $Q = A^\top A / M^2$ (Appendix A), and the stated fractions are $1 - b$. An independent implementation rebuilt every Q from the certificate files and computed the exact minimum b by a raw scan of all one-vertex extensions of its own 6-vertex representatives (29 092 719, 40 527 893 and 10 058 620 admissible extensions for J_4 , $K_5^<$ and K_5^{4-} , and from 5 635 379 for C_5 to 163 577 855 for $K_7^{(4)}$ for the others), and again over the class lists (except for K_6^{4-}), with the same group maxima. $\square$

F	lower bound	previous upper bound	this paper
J_4	$16/57 > 0.280701$ (Prop. 28)	0.28079696 [2]	0.2807037
$K_5^<$	$1/3$ [2]	0.3402107 [2]	0.3336455
$K_5^=$	$2909/8127 > 0.357942$ [2]	0.388725 [2]	0.3844257
K_5^{3-}	$2909/8127 > 0.357942$ [2]	0.41961743 [2]	0.4059302
C_5	> 0.251940 [2]	0.25310457 [2]	0.2523050
$K_6^{(3)}$	> 0.734799 [2]	0.7535963 [2]	0.7423183
J_5	$36/91 > 0.395604$ (Prop. 28)	0.4427457 [2]	—
K_5^{4-}	$1/5$ (construction of [36])	$1/4$ (derivable)	0.2040329
K_6^{4-}	> 0.501220 (Rem. 32)	< 0.852398 (via π)	0.6120380
$K_6^{(4)}$	> 0.719219 (Rem. 32)	< 0.894570 (via π)	0.7437531
$K_7^{(4)}$	> 0.834439 (via π)	< 0.944386 (via π)	0.8771552

Table 7: Codegree-squared densities: upper bounds from plain certificates on seven vertices (rounded up) against the literature (upper bounds of [2] as printed there, other upper bounds rounded up, lower bounds rounded down). For the 4-graphs, “via π ” uses $\pi(F)^2 \leq \sigma(F) \leq \pi(F)$ (Cauchy–Schwarz, and $\deg_G(T) \leq n - 3$) with the previous bounds of Table 1; with the bounds of Theorem 2 the upper ones become 0.780548, 0.860173 and 0.935238. The lower bound for $K_7^{(4)}$ is $(4677/5120)^2$, from [34, Prop. 6.1]. For C_5 the construction of [2, Section 3.17 of arXiv v3] has the edges xyz with $x, y \in B$, $z \in A$, $|B| = bn$, iterated inside A (the part sizes and the part of the iteration are interchanged there; the value is right). For J_5 our seven-vertex run was stopped before it reached the bound 0.4427457 of [2]; we claim no bound.

The lower bound $1/5$ for $\sigma(K_5^{4-})$ in Table 7 comes from Sidorenko’s doubling construction [36]: with $h = n/2$, its codegree-squared sums satisfy $c_{L+1} = 2c_L + 2\binom{h}{2}h(h-1)^2$, so $\text{co}_2 \sim n^5/30$. For K_5^{4-} we found no published upper bound for σ . The bound $1/4$ is easy: if G is K_5^{4-} -free, then for every edge e and every vertex $x \notin e$ at most two of the four sets $T \cup \{x\}$ ($T \subset e$, $|T| = 3$) are edges, so $\sum_{T \subset e} \deg_G(T) \leq 2n - 4$ and $\text{co}_2(G) \leq (2n - 4)e(G)$ (de Caen’s double count, as in the proof of [2, Theorem 8.3 of arXiv v3]); with $\pi(K_5^{4-}) \leq 1/2$ [23, Cor. 3.9] this gives $\sigma(K_5^{4-}) \leq 1/4$, which also follows from a one-block certificate on five vertices. Except for the interval in Remark 31, these bounds are not claimed to be seven-vertex optima: the remaining gaps neither show nor rule out that seven vertices reach the lower bounds.

Remark 31. Seven vertices cannot reach $16/57$. An exact dual point on seven vertices, as in Section 5, is a probability vector y on 757 J_4 -free 7-vertex 3-graphs, with weights in $2^{-110}\mathbb{Z}$, whose moment matrices are all positive semidefinite (checked exactly, for all labelled types and all flag sizes with $2m - s \leq 7$). As in Lemma 20, every plain seven-vertex certificate for J_4 then has value $b \leq V = \sum_H y_H(1 - \gamma(H))$, with $1 - V = 0.2807032753\dots$, so it proves at best $\sigma(J_4) \leq 1 - V$: the plain seven-vertex method cannot prove any bound below 0.280703275315, and in particular misses $16/57$ by more than $1.5 \cdot 10^{-6}$. With Theorem 30, the plain seven-vertex optimum lies in $(0.280703275315, 0.280703690307]$. The obstruction is structural: for the law ϕ of the 7-vertex subgraph of the construction (93 classes, $\mathbb{E}_\phi(1 - \gamma) = 41/57$), a certificate of value $41/57$ would need $\mathbb{E}_\phi c_Q = 0$ for every block, and an exact rational witness on the support of ϕ shows that such certificates prove at best $\sigma(J_4) \leq 16/57 + \delta$, $\delta = 316891/14980607589 \approx 2.1 \cdot 10^{-5}$. As for Theorem 24, none of this covers stationarity constraints or more than seven vertices. For $K_5^<$ the run stopped before convergence, so it neither shows nor rules out that seven vertices reach $1/3$.

Remark 32. Two further lower bounds for 4-graphs: $\sigma(K_6^{4-}) > 0.5012201$, above the lower bound $\sigma(K_6^{4-}) \geq \sigma(K_5^{(4)}) = 31/64$, by a three-part pattern construction (parts of relative sizes 0.36827, 0.355115, 0.276615, whose edges are the 4-sets of patterns $(3, 0, 1)$, $(2, 2, 0)$, $(2, 1, 1)$, $(1, 2, 1)$, $(1, 1, 2)$, $(1, 0, 3)$, $(0, 3, 1)$; every six vertices span at most 13 edges); and

$\sigma(K_6^{(4)}) > 0.7192198$ for the complement of Sidorenko's construction [34, Ex. 3.4] with equal Giraud halves and part weights optimised for σ .

7 Applications

7.1 Lottery numbers

An (n, k, r, p) -lottery system is a family of k -subsets (tickets) of an n -set such that every p -subset (draw) meets some ticket in at least r points; $L(n, k, r, p)$ is its least size. We follow Sidorenko's order of arguments [35]; the lotto tables [21] write $L(n, k, p, r)$, so their $L(n, 6, 6, 4)$ is $L(n, 6, 4, 6)$ here. Sidorenko proved that $\ell(k, r, p) = \lim_n L(n, k, r, p) / \binom{n}{r}$ exists [35].

Lemma 33 (Shadow bound [3, Lemma 4.1.1], [35, eq. (2)]). *The r -subsets of the tickets of an (n, k, r, p) -lottery system form a Turán (n, p, r) -system. Hence $L(n, k, r, p) \geq T(n, p, r) / \binom{k}{r}$ and $\ell(k, r, p) \geq t(p, r) / \binom{k}{r}$.*

Proof. A draw that meets a ticket in at least r points contains an r -subset of that ticket, and a ticket has $\binom{k}{r}$ such subsets. $\square$

Corollary 34. *For every $k \geq 4$,*

$$\begin{aligned} \binom{k}{4} \ell(k, 4, 5) &> 0.308400, & \binom{k}{4} \ell(k, 4, 6) &> 0.139827, \\ \binom{k}{4} \ell(k, 4, 7) &> 0.064762. \end{aligned}$$

In particular $L(n, 6, 4, 6) \geq (1 - o(1)) \binom{n}{4} / 107.28$.

Table 8 gives the cases $5 \leq k, p \leq 7$. These are asymptotic statements; they do not improve entries of the finite tables [21] directly. The previous lower bounds are the same shadow bound with the previous values of $t(p, 4)$. The upper bounds for $(6, 4, 5)$, $(7, 4, 5)$ and $(7, 4, 6)$ are those of Proposition 44 (Section 7.3).

$(k, 4, p)$	this paper	previous	upper bound	source of the upper bound
$(5, 4, 5)$	16.22	18.92	12.48	[27, Cor. 5.6]
$(6, 4, 5)$	48.64	56.74	34.28	Prop. 44(a)
$(7, 4, 5)$	113.49	132.39	78.23	Prop. 44(b)
$(5, 4, 6)$	35.76	47.43	20.44	[27, Thm 5.9]
$(6, 4, 6)$	107.28	142.28	58.24	[27, Section 6], from [26]
$(7, 4, 6)$	250.31	331.98	124.05	Prop. 44(c)
$(5, 4, 7)$	77.21	89.91	40	[27, eq. (12)]
$(6, 4, 7)$	231.62	269.72	120	[27, eq. (12)]
$(7, 4, 7)$	540.44	629.33	280	[27, eq. (12)]

Table 8: Asymptotic bounds on $L(n, k, 4, p)$, written as $\binom{n}{4}/x$ up to a factor $1 \pm o(1)$; the table gives x , rounded up for the lower bounds and down for the upper bounds. Previous lower bounds use $t(5, 4) > 0.264386$, $t(6, 4) > 0.105430$ and $t(7, 4) > 0.055614$, by repeated averaging from Sidorenko's table values [34] as in Table 1. Montecalvo's $\delta(k, t, p)$ is $\binom{p}{t}$ times our normalised constant $\binom{k}{t} \ell(k, t, p)$; the value for $(6, 4, 6)$, $\delta < 3.863$, is quoted in [27] from [26]. The previous upper bounds for the rows of Proposition 44 were 33.21 for $(6, 4, 5)$ ($\delta \leq 2.258$, quoted in [27, Section 5.2] from [26]), and 74.60 and 122.84 for $(7, 4, 5)$ and $(7, 4, 6)$ [27, Thm 5.12], a theorem that assumes that the limits exist, which Sidorenko proved [35].

7.2 Giraud systems are expensive to cover

Lemma 33 gives $\binom{k}{4}\ell(k, 4, 5) \geq t(5, 4)$, and we do not know whether equality can hold for some $k \geq 5$. If Sidorenko's conjecture $t(5, 4) = 5/16$ holds, equality would require lottery systems whose shadows are near-extremal Turán systems, and the best known constructions of Turán $(n, 5, 4)$ -systems are Giraud's [33, Section 8], [34, Section 2]. We show that these are far from being shadows of cheap lottery systems.

Recall the Giraud systems of Definition 3 and their density bound in Lemma 10. For a 4-graph Γ on $[n]$ and $k \geq 5$ let $\tau_k(\Gamma)$ be the fractional cover number of $E(\Gamma)$ by k -subsets of $[n]$, the least $\sum_K w_K$ over weights $w \geq 0$ on k -sets with $\sum_{K \supseteq T} w_K \geq 1$ for every edge T . Since a k -set contains $\binom{k}{4}$ quadruples, $\binom{k}{4}\tau_k(\Gamma) \geq e(\Gamma)$.

Theorem 35. *For every $k \geq 5$ and every Giraud system $\Gamma = \Gamma(A, B, M)$ on n vertices, with $s = |A||B|/n^2$,*

$$\binom{k}{4} \frac{\tau_k(\Gamma)}{\binom{n}{4}} \geq 1 - 4s + 7s^2 - O(1/n) \geq \frac{7}{16} - O(1/n),$$

uniformly in M and the part sizes. The density of Γ can be as small as $5/16 + o(1)$.

Lemma 36. *Let $k \geq 5$, $i + j = k$ and $i, j \geq 0$. Then $\binom{i}{4} + \binom{j}{4} + \frac{5}{3}\binom{i}{2}\binom{j}{2} \leq \binom{k}{4}$.*

Proof. By Vandermonde, $\binom{k}{4} = \binom{i}{4} + \binom{i}{3}j + \binom{i}{2}\binom{j}{2} + i\binom{j}{3} + \binom{j}{4}$, so the claim is $\frac{2}{3}\binom{i}{2}\binom{j}{2} \leq \binom{i}{3}j + i\binom{j}{3}$. It is trivial if $i \leq 1$ or $j \leq 1$. Otherwise divide by $ij/6$ and put $a = i - 1$, $b = j - 1 \geq 1$; the claim becomes $ab \leq a(a - 1) + b(b - 1)$, that is $(a - b)^2 + (a - 1)(b - 1) - 1 \geq 0$. If $a \neq b$ the first term is at least 1. If $a = b$, then $2a = k - 2 \geq 3$, so $a \geq 2$ and $(a - 1)^2 \geq 1$. $\square$

Proof of Theorem 35. If a part has fewer than two points, then $s = O(1/n)$, Γ contains all but $O(n^3)$ quadruples, and the claim follows from $\binom{k}{4}\tau_k(\Gamma) \geq e(\Gamma)$. Otherwise, by LP duality, $\tau_k(\Gamma) \geq \sum_T y_T$ for every $y \geq 0$ on $E(\Gamma)$ with $\sum_{T \subseteq K} y_T \leq 1$ for every k -set K . Give weight $y_I = 1/\binom{k}{4}$ to the internal edges and $y_X = \frac{5}{3}/\binom{k}{4}$ to the cross edges. A k -set with i points in A and j in B contains exactly $\binom{i}{4} + \binom{j}{4}$ internal edges and at most $\binom{i}{2}\binom{j}{2}$ cross edges, whatever M is; so Lemma 36 says that y is feasible. Hence

$$\binom{k}{4}\tau_k(\Gamma) \geq \binom{|A|}{4} + \binom{|B|}{4} + \frac{5}{3}f\binom{|A|}{2}\binom{|B|}{2},$$

and Lemma 10 gives $1 - 4s + 2s^2 + 10s^2f - O(1/n) \geq 1 - 4s + 7s^2 - O(1/n)$ after division by $\binom{n}{4}$; the error term from f is $O(1/n)$ because $\max(|A|, |B|) \geq n/2$. Finally $1 - 4s + 7s^2$ is decreasing on $[0, 1/4]$, with value $7/16$ at $s = 1/4$. $\square$

Remark 37. For $k = 5, 6$ no other choice of the two weights gives more. Optimising the two weights for each k at the extremal members ($s = 1/4$, $f = 1/2$) gives the exact values $7/16, 7/16, 23/48, 23/48, 1/2, 1/2, 41/80, 41/80$ for $k = 5, \dots, 12$; over all part sizes the minimum is the same for $k \leq 10$ and $21/41$ (at $s = 10/41$) for $k = 11, 12$. For $k = 5$ the proof is one line: a five-set containing a cross edge has at most three edges, all cross, so $\tau_5 \geq |E_{\text{int}}|/5 + |E_{\text{cross}}|/3$. For $k = 6$ the bound $7/16$ of Theorem 35 is sharp. The weights in the proof of Proposition 44(a) below are a fractional cover by 6-sets of the edges of Γ_d , which is Giraud's construction in the sense of this section (equal parts, $f \rightarrow 1/2$), so $\binom{6}{4}\tau_6(\Gamma_d)/\binom{n}{4} \leq 15W_N/\binom{n}{4} \rightarrow 7/16$; by Lemma 42 there is even an integral cover of the same asymptotic size, which is an $(n, 6, 4, 5)$ -lottery system.

For 4-graphs Γ, Γ' on $[n]$ write $\Gamma \setminus \Gamma'$ for the set difference of the edge sets. The shadow ∂G of a lottery system G is the 4-graph of the 4-subsets of its tickets.

Corollary 38. *Let G be an $(n, k, 4, 5)$ -lottery system and Γ a Giraud system on the same n points with $|\Gamma \setminus \partial G| \leq \eta n^4$. Then $\binom{k}{4}|G|/\binom{n}{4} \geq 7/16 - 24\binom{k}{4}\eta - o(1)$.*

Proof. The tickets of G , together with one k -set through each edge of $\Gamma \setminus \partial G$, cover every edge of Γ . So $\tau_k(\Gamma) \leq |G| + \eta n^4$, and $n^4/\binom{n}{4} \rightarrow 24$. $\square$

Corollary 39. *If G_n are $(n, 5, 4, 5)$ -lottery systems of size $L(n, 5, 4, 5)$, then every Giraud system Γ on the points of G_n has $|\Gamma \setminus \partial G_n| \geq (1/3240 - o(1))n^4 = (1/135 - o(1))\binom{n}{4}$.*

Proof. Montecalvo [27, Cor. 5.6] gives $5L(n, 5, 4, 5)/\binom{n}{4} \leq 865/2160 + o(1)$. With $k = 5$, Corollary 38 forces $120\eta \geq 7/16 - 865/2160 - o(1) = 1/27 - o(1)$. $\square$

So optimal $(n, 5, 4, 5)$ -lottery systems do not have shadows close to Giraud systems. For $k = 6$ Theorem 35 gives no such statement: the lottery systems in the proof of Proposition 44(a) (Section 7.3) cover every edge of Γ_d and attain its bound $7/16$. Instead, Corollary 38 says that an $(n, 6, 4, 5)$ -lottery system G with $\binom{6}{4}|G|/\binom{n}{4} \leq 7/16 - \varepsilon$ has $|\Gamma \setminus \partial G| \geq (\varepsilon/360 - o(1))n^4$ for every Giraud system Γ , so any improvement of Proposition 44(a) must have a shadow far from every Giraud system. With a stability hypothesis, Corollary 38 gives a strict inequality in the shadow bound for every k (Corollary 41).

Hypothesis 40 (S_G). For every $\varepsilon > 0$ there are $\delta > 0$ and n_0 such that every Turán $(n, 5, 4)$ -system with $n \geq n_0$ and density at most $5/16 + \delta$ satisfies $|\Gamma \setminus G| \leq \varepsilon n^4$ for some Giraud system Γ on the same vertex set.

Hypothesis S_G implies Sidorenko's conjecture $t(5, 4) = 5/16$, because a Giraud system has density at least $5/16 - o(1)$. It is a stability form of that conjecture. Markström conjectured that for $n \geq 12$ the only extremal $K_5^{(4)}$ -free 4-graphs are those of Giraud's construction and its generalisation [22, Conjecture 4.2], and he wrote that he expects "a stability version of Conjecture 4.2 ... showing that the near-extremal hypergraphs are also close to Giraud's construction" [22, Section 4.1]. Hypothesis S_G is a version of this, in complement form, that allows any Giraud system.

Corollary 41. *If Hypothesis S_G holds, then $\binom{k}{4}\ell(k, 4, 5) > 5/16 \geq t(5, 4)$ for every $k \geq 5$: the shadow bound is not tight for $(k, 4, 5)$.*

Proof. Put $\varepsilon = 1/(384\binom{k}{4})$ and let δ be given by S_G . Let G be an optimal lottery system. If $d(\partial G) > 5/16 + \delta$, then $\binom{k}{4}|G|/\binom{n}{4} \geq d(\partial G) > 5/16 + \delta$. Otherwise ∂G is a Turán $(n, 5, 4)$ -system (Lemma 33) of density at most $5/16 + \delta$, and Corollary 38 gives $\binom{k}{4}|G|/\binom{n}{4} \geq 7/16 - 24\binom{k}{4}\varepsilon - o(1) = 3/8 - o(1)$. In both cases the limit exceeds $5/16$, and $t(5, 4) \leq 5/16$ by Giraud's construction. $\square$

7.3 Upper bounds for lottery numbers

We improve three upper bounds of Table 8, Montecalvo's normalised constants 0.4516 , $38/81$ and $7385/25920$ [27], with fractional covers made integral by Kahn's fractional form [18] of the Frankl–Rödl–Pippenger theorem [12, 31]. For a hypergraph X on a finite set V , a map $\varphi: X \rightarrow \mathbb{R}_{\geq 0}$ is a *fractional cover* if $\sum_{A \ni v} \varphi(A) \geq 1$ for every $v \in V$; let $\varphi(X) = \sum_{A \in X} \varphi(A)$ and $\alpha(\varphi) = \max_{u \neq v} \sum_{A \ni u, v} \varphi(A)$, and let $\rho(X)$ be the least number of edges of X whose union is V . Kahn [18, Theorem 1.2(b)] proved: for fixed h , if every edge of X has at most h elements and φ is a fractional cover, then $\rho(X) \leq (1 + o(1))\varphi(X)$ as $\alpha(\varphi) \rightarrow 0$, uniformly in X . (He derives it, for $\varphi(X) \rightarrow \infty$, from his Theorem 1.4 on fractional matchings, applied to a *fractional tiling*, where $\sum_{A \ni v} \varphi(A) = 1$ for every v . Moving, vertex by vertex, the excess weight at v from edges $A \ni v$ to $A \setminus \{v\}$ (deleting it if $A = \{v\}$) turns a fractional cover into a tiling of the down-closure of X with no larger $\varphi(X)$ or $\alpha(\varphi)$, and a cover of the down-closure gives one of X of the same size. In part (a) of Proposition 44, φ is already a tiling.)

Lemma 42. Fix $k \geq r$. For n in an infinite set let G_n be an r -graph on $[n]$ with $e(G_n) \rightarrow \infty$, and $w_n \geq 0$ weights on the k -subsets of $[n]$ with $\sum_{K \supseteq R} w_n(K) \geq 1$ for every edge R of G_n and $\alpha_n = \max_{R \neq R'} \sum_{K \supseteq R \cup R'} w_n(K) \rightarrow 0$, the maximum over distinct edges R, R' of G_n . Then there is a family of at most $(1 + o(1)) \sum_K w_n(K)$ k -sets such that every edge of G_n lies in one of them. If G_n is a Turán (n, p, r) -system, this family is an (n, k, r, p) -lottery system.

Proof. Let X_n be the hypergraph on $E(G_n)$ whose edges are the nonempty sets $E_K = \{R \in E(G_n) : R \subseteq K\}$ with $w_n(K) > 0$, and put $\varphi(E) = \sum_{K: E_K=E} w_n(K)$. Its edges have at most $\binom{k}{r}$ elements, φ is a fractional cover, $\alpha(\varphi) = \alpha_n$ and $e(G_n)/\binom{k}{r} \leq \varphi(X_n) \leq \sum_K w_n(K)$. Kahn's theorem gives a cover of X_n by $(1 + o(1)) \varphi(X_n)$ sets E_K ; take their k -sets K . A p -set contains an edge of a Turán (n, p, r) -system, and hence meets the k -set containing that edge in at least r points. $\square$

For type patterns fix $m \geq 1$ and weights $x \in (0, 1]^m$ with $\sum_i x_i = 1$. A *type* is a vector $e \in \mathbb{Z}_{\geq 0}^m$ of size $|e| = \sum_i e_i$; for a partition $[n] = V_1 \cup \dots \cup V_m$, the type of a set S is $(|S \cap V_i|)_{i \leq m}$. For types e of size r and B of size k let $a(e, B) = \prod_i \binom{B_i}{e_i}$, the number of r -subsets of type e of a k -set of type B , and $\mu_e(x) = r! \prod_i x_i^{e_i} / e_i!$. We write $P \geq e$ coordinatewise.

Lemma 43. Let $k \geq r$, $p \geq r$ and $q \geq r$, let $I \subseteq [m]$, and let $\mathcal{T}$ be a nonempty set of types of size r such that

- (i) every type P of size p with $P_i < q$ for all $i \in I$ satisfies $P \geq e$ for some $e \in \mathcal{T}$, and
- (ii) some $y = (y_B) \geq 0$, indexed by the types B of size k , satisfies $\sum_B a(e, B) y_B \geq \mu_e(x)$ for every $e \in \mathcal{T}$.

Then $\ell(k, r, p) \leq \sum_B y_B + \ell(k, r, q) \sum_{i \in I} x_i^r$.

Proof. Split $[n]$ into parts V_i with $|V_i| = x_i n + O(1)$ and let G_n consist of the r -sets with type in $\mathcal{T}$. There are $N_e = \prod_i \binom{|V_i|}{e_i} = (1 + O(1/n)) \mu_e(x) \binom{n}{r}$ r -sets of type e and $N_B = \prod_i \binom{|V_i|}{B_i} = \Theta(n^k)$ k -sets of type B , and an r -set of type e lies in $a(e, B) N_B / N_e$ of the latter. Give each k -set of type B the weight $\lambda y_B \binom{n}{r} / N_B$, where $\lambda = \max_{e \in \mathcal{T}} N_e / (\mu_e(x) \binom{n}{r}) = 1 + O(1/n)$. By (ii) every $R \in G_n$ of type e gets total weight $\lambda \binom{n}{r} N_e^{-1} \sum_B a(e, B) y_B \geq 1$; the total weight is $\lambda \binom{n}{r} \sum_B y_B$; and $\alpha_n = O(1/n)$, because two distinct r -sets span at least $r + 1$ points, which lie in $O(n^{k-r-1})$ k -sets, each of weight $O(n^{r-k})$. Lemma 42 gives $(1 + o(1)) \binom{n}{r} \sum_B y_B$ k -sets that contain all members of G_n . Add, for each $i \in I$, an optimal $(|V_i|, k, r, q)$ -lottery system on V_i ; it has $(\ell(k, r, q) x_i^r + o(1)) \binom{n}{r}$ tickets, since $\ell(k, r, q)$ is a limit [35]. Every p -set P meets some ticket in at least r points: if $|P \cap V_i| \geq q$ for some $i \in I$, a ticket of the system on V_i ; otherwise, by (i), P contains an r -set with type in $\mathcal{T}$, which lies in one of the k -sets above. Divide by $\binom{n}{r}$ and let $n \rightarrow \infty$. $\square$

Proposition 44. (a) $\binom{6}{4} \ell(6, 4, 5) \leq 7/16$; that is, $L(n, 6, 4, 5) \leq (1 + o(1)) 7 \binom{n}{4} / 240 \leq (1 + o(1)) \binom{n}{4} / 34.28$.

(b) $\binom{7}{4} \ell(7, 4, 5) \leq c_{745}$, where

$$c_{745} = \frac{3467292853891919529774577}{775000000000000000000000000000} < 0.447393.$$

(c) $\binom{7}{4} \ell(7, 4, 6) \leq (107 + 3c_{745}) / 384 < 0.282142$.

Proof. (a) Let $d \geq 4$, $N = 2^d$ and $n = 2N$, let A and B be disjoint copies of $\mathbb{F}_2^d$, and let Γ_d be the Giraud system $\Gamma(A, B, M)$ of Definition 3 with $M_{ab} = \langle a, b \rangle$, the standard bilinear form over $\mathbb{F}_2$. The minor of $\{a, a', b, b'\}$ is $\langle a + a', b + b' \rangle$, so for fixed $a \neq a'$ a fraction

$(N/2 - 1)/(N - 1)$ of the pairs $\{b, b'\}$ give cross edges and $f \rightarrow 1/2$; by Lemma 10, Γ_d is a Turán $(n, 5, 4)$ -system with equal parts. Call a 6-set *good* if it lies in A or in B , or if it is a set $\{a_1, a_2, a_3, b_1, b_2, b_3\}$ whose nine cross 4-subsets are edges. With $u = a_1 + a_2$, $v = a_1 + a_3$, $w = b_1 + b_2$ and $z = b_1 + b_3$, the differences inside the two triples are $u, v, u + v$ and $w, z, w + z$, so by bilinearity the set is good iff $\langle u, w \rangle = \langle u, z \rangle = \langle v, w \rangle = \langle v, z \rangle = 0$. Given a cross edge $\{a_1, a_2, b_1, b_2\}$, so that $u, w \neq 0$ and $\langle u, w \rangle = 0$, the good sets containing it correspond to the pairs (v, z) with $v \in w^\perp \setminus \{0, u\}$, which leaves $N/2 - 2$ choices because $u \in w^\perp$, and $z \in \{u, v\}^\perp \setminus \{0, w\}$, which leaves $N/4 - 2$ choices because u, v are linearly independent and $w \in \{u, v\}^\perp$. So every cross edge lies in exactly $c_N = (N/2 - 2)(N/4 - 2)$ good sets, and every internal edge in $\binom{N-4}{2}$ good sets inside its part. The weights $1/c_N$ on the good sets of type $(3, 3)$ and $1/\binom{N-4}{2}$ on the other good sets give every edge of Γ_d total weight exactly 1, and since a good set contains 9 cross or 15 internal edges, the total weight is

$$W_N = \frac{|E_{\text{cross}}|}{9} + \frac{2\binom{N}{4}}{15}, \quad |E_{\text{cross}}| = \binom{N}{2} \left(\frac{N}{2} - 1\right) \frac{N}{2},$$

so that $15W_N/\binom{n}{4} \rightarrow 15(\frac{1}{72} + \frac{1}{180})\frac{3}{2} = \frac{7}{16}$. Two distinct edges in a common good set span at least five of its points, which leaves at most N choices for the sixth, and no good set contains both an internal and a cross edge; so $\alpha_n \leq N \max(1/c_N, 1/\binom{N-4}{2}) = O(1/N)$. Lemma 42 gives $L(n, 6, 4, 5) \leq (1 + o(1))W_N$ along $n = 2^{d+1}$, and $\ell(6, 4, 5)$ is a limit [35].

(b) This is Lemma 43 with $m = 5$ and $I = \emptyset$ (so q plays no role), the part weights $x_1 = 0.317221$, $x_2 = x_4 = 0.126888$, $x_3 = 0.302114$ and $x_5 = 0.126889$, a set $\mathcal{T}$ of 38 types (the five types with four points in one part, ten of shape $(3, 1)$, ten of shape $(2, 2)$, twelve of shape $(2, 1, 1)$ and one of shape $(1, 1, 1, 1)$) and 27 positive rational values y_B with $\binom{7}{4} \sum_B y_B = c_{745}$, listed in the data package [25]. Conditions (i) (on the 126 types of size 5) and (ii) (38 rational inequalities, 27 of them tight) were checked in exact arithmetic by two independent programs, and (i) also by brute force on a blow-up with five points in each part.

(c) This is Lemma 43 with $m = 4$, $x_i = 1/4$, $q = 5$ and $I = \{1, 2\}$. Write $i \rightarrow j$ for an ordered pair of parts and use the six arcs $1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4, 4 \rightarrow 1, 1 \rightarrow 3, 2 \rightarrow 4$, one on each pair of parts. $\mathcal{T}$ consists of the types (4) inside parts 3 and 4, the six types $(2, 2)$, and for each arc $i \rightarrow j$ the type with three points in part i and one in part j . Put $y_B = 1/13440$ for the 7-sets inside part 3 and for those inside part 4, and $y_B = 1/768$ for the six block types with four points in part i and three in part j , one for each arc $i \rightarrow j$. For (i), let P have size 6 and $P_1, P_2 \leq 4$. If $P_3 \geq 4$, or $P_4 \geq 4$, or two parts have at least two points each, P is at least a type of $\mathcal{T}$. Otherwise some part i has $P_i \geq 3$ and every other part at most one point. If $P_i = 3$, the other three parts have a point each and i is the tail of an arc; if $P_i = 4$, then $i \in \{1, 2\}$, two of the other parts have a point, and i is the tail of two arcs ($1 \rightarrow 2, 3$, resp. $2 \rightarrow 3, 4$), one of which ends in such a part. For (ii), at $x_i = 1/4$ we have $\mu_e = 1/256, 3/128$ and $1/64$ for the types (4), $(2, 2)$ and $(3, 1)$. The 4-sets inside part 3 occur only in the 7-sets inside part 3, 35 in each, and in the blocks of the arc $3 \rightarrow 4$, one in each: $35/13440 + 1/768 = 1/256$; the same holds for part 4 with the arc $4 \rightarrow 1$. A type $(2, 2)$ occurs only in the blocks of the arc on its pair, $\binom{4}{2}\binom{3}{2} = 18$ times in each, and the type $(3, 1)$ of an arc only in the blocks of that arc, $\binom{4}{3}\binom{3}{1} = 12$ times in each: $18/768 = 3/128$ and $12/768 = 1/64$. So all 14 rows are tight, and $\binom{7}{4} \sum_B y_B = 35(2/13440 + 6/768) = 107/384$. The second term of Lemma 43, multiplied by $\binom{7}{4}$, is $\binom{7}{4}\ell(7, 4, 5)(x_1^4 + x_2^4) \leq c_{745}/128$ by (b), and $107/384 + c_{745}/128 = (107 + 3c_{745})/384$. $\square$

The pattern in (c) has the shape of Montecalvo's bound [27, Theorem 5.10(3)]. With his value $38/81$ for $(7, 4, 5)$ in place of c_{745} it gives $(107 + 3 \cdot \frac{38}{81})/384$, exactly $1/384$ below his $7385/25920$; this gain comes from the rows of type (4), to which the blocks with four points in parts 3 and 4 also contribute.

7.4 Covering insertion codes

A set $C \subseteq [n]^r$ is a *covering $(k-r)$ -insertion code* if every word in $[n]^k$ contains a member of C as a subsequence; let $S(n, k, r)$ be its least size and $s(k, r) = \lim_n S(n, k, r)/n^r$. Pikhurko, Verbitsky and Zhukovskii proved $s(k, r) = t(k, r)$ [30, Theorem 18 of arXiv v3].

Corollary 45. $s(5, 4) > 0.308400$, $s(6, 4) > 0.139827$, $s(7, 4) > 0.064762$, $s(6, 5) > 0.232599$ and $s(7, 5) > 0.072082$.

8 Data and reproducibility

The data package [25] contains the exact certificates, independent verifiers, search code and file checksums. Code is under the MIT licence; data are under CC BY 4.0. Verification does not require the search code or the stored class lists: the independent verifier generates its own six-vertex representatives and scans all raw extensions, as in Section 3.5.

The package documents each certificate and the corresponding check. For the five bounds of Theorem 2, the independent C evaluator takes about ten minutes to three and a half hours per raw scan on two cores. The `verifier/verify_all.sh -fast` and `-raw` all checks were run on clean cloud machines before release; all passed.

AI disclosure

AI coding agents, built on several models (including Claude and GPT), assisted the implementation, the computations, the proofs and the writing of this paper. The search code that found the certificates and the verifiers that re-checked them were written in separate agent sessions that did not share code, and every result was checked by a separate agent session before it entered the paper. The cutting-plane and column-generation scheme follows Jeong et al. [17], in an independent implementation that does not use their code. The author ran the fast checks of the data package (`verify_all.sh -fast`); all its checks, including the exhaustive scans, were also run on clean cloud machines.

A Exact bounds for the further results

The further codegree-squared certificates have $Q = A^T A / M^2$; the matrices and scales are specified in the data package [25].

The exact bounds for the remaining entries of Table 7 are

$$\begin{aligned} \sigma(K_5^-) &\leq \frac{66572174747207}{173173081374720}, & \sigma(K_5^{3-}) &\leq \frac{781068653059}{1924145348608}, & \sigma(C_5) &\leq \frac{2167282889}{8589934592}, \\ \sigma(K_6^{(3)}) &\leq \frac{85699693910797}{115448720916480}, & \sigma(K_6^{4-}) &\leq \frac{565272027609637}{923589767331840}, \\ \sigma(K_6^{(4)}) &\leq \frac{408882571413}{549755813888}, & \sigma(K_7^{(4)}) &\leq \frac{67510962453727}{76965813944320}. \end{aligned}$$

The exact bounds of Theorem 27 (Table 6) are

$$\begin{aligned} t_3(5, 4) &\geq \frac{3298534811415}{4398046511104}, & t_3(6, 4) &\geq \frac{11166973764957}{38482906972160}, & t_4(6, 4) &\geq \frac{391547931101}{1099511627776}, \\ t_5(6, 4) &\geq \frac{64873027783677}{153931627888640}, & t_6(6, 4) &\geq \frac{612132591153137}{1231453023109120}, & t_9(6, 4) &\geq \frac{217134104143267}{307863255777280}, \\ t_{11}(6, 4) &\geq \frac{24267758577011}{28862180229120}, & t_2(7, 4) &\geq \frac{111867473425}{1099511627776}, & t_3(7, 4) &\geq \frac{4043518541347}{30786325577728}, \\ & & t_4(7, 4) &\geq \frac{11199502214035}{69269232549888}. \end{aligned}$$

The exact bounds of Theorem 26 are

$$\begin{aligned} t(6, 5) &\geq \frac{5410603883828909241569858123831800227}{23261489926236027775816623554906030080} > 0.232599197256, \\ t_3(6, 5) &\geq \frac{28392597905796184739036021237749789573}{46522979852472055551633247109812060160} > 0.610291902965, \\ t(7, 5) &\geq \frac{1006048490611906127582298374928111023}{13956893955741616665489974132943618048} > 0.072082548868, \end{aligned}$$

where $t_3(6, 5)$ is the least density of 5-graphs in which every six vertices span at least three edges, so that $\pi(H_4^5) = 1 - t_3(6, 5)$.

B Notation

$T_\lambda(n, p, r), T(n, p, r)$	least number of edges of an r -graph on n vertices in which every p -set spans at least λ edges; $T = T_1$ (Definition 1)
$t_\lambda(p, r), t(p, r)$	$\lim_n T_\lambda(n, p, r)/\binom{n}{r}$; $t = t_1$
$\pi(F)$	Turán density; $\pi(K_p^{(r)}) = 1 - t(p, r)$, $\pi(H_k^r) = 1 - t_{r+2-k}(r+1, r)$, $\pi(K_6^{4-}) = 1 - t_2(6, 4)$
$d(H), c_Q(H), b$	edge density, sum-of-squares term and value of a certificate (Definition 6)
$\text{co}_2(G), \sigma(F)$	codegree-squared sum $\sum_T \deg_G(T)^2$ and codegree-squared density [2]
C_0	$21894/\mu_0 < 7.81 \cdot 10^8$, the constant of Theorem 17
L_v ; locally Giraud	link $\{R : R \cup \{v\} \in E(H)\}$; every 7-vertex induced subgraph is a Giraud system (Section 4.3)
$\mu_0, q(H), \xi(H)$	margin of the certificate of Theorem 4 outside Giraud's classes, fraction of non-Giraud 7-sets, and $\kappa(H) - 33/64 + 68/(n-7)$ (Proposition 12)
$\gamma(G), \kappa(H)$	probability that $T \cup \{x\}, T \cup \{y\}$ are both edges; $\kappa = 2d - \gamma$ (Lemma 11)
$L(n, k, r, p), \ell(k, r, p)$	lottery number (tickets of size k , match r , draws of size p) and $\lim_n L(n, k, r, p)/\binom{n}{r}$; the lotto tables write $L(n, k, r, p)$
$\binom{k}{r} \ell(k, r, p)$	normalised lottery constant; the shadow bound says it is at least $t(p, r)$
$\delta(k, t, p)$	Montecalvo's constant, $\binom{p}{t}$ times the normalised lottery constant
$s(k, r)$	limit density of covering $(k-r)$ -insertion codes; $s(k, r) = t(k, r)$ [30]
$\tau_k(\Gamma)$	fractional cover number of the edges of Γ by k -sets (Section 7.2)

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